

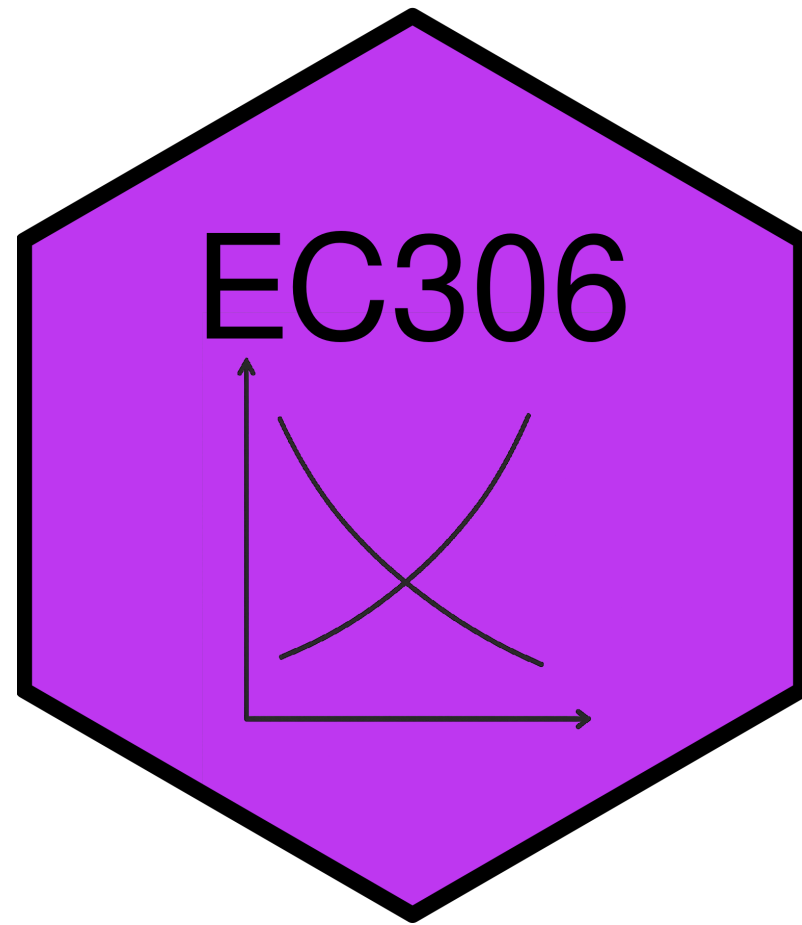
Labour Supply 1: Individual Attachment to the Labour Market

EC306- Wages and Employment

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Goals of This Section

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- Introduce common labour market concepts
- Build the labour supply model
- Derive labour supply curve
- Examine effects of labour market policies within that model

Key Labour Force Concepts

Introduction

- The labour market involves several individual decisions about work
 - It is not simply whether to work or not
- In the news you hear about statistics that quantify these decisions
- In this section we cover the most common ones

Labour Force Participation

- The first decision is whether to make yourself *available* for work
- **Labour Force:** people in the eligible population who participate in labour market activities as either employed or unemployed
 - A measure of the people that could work or are working
- Eligible population includes
 - Population 15 years and over
 - Non-institutionalized
 - People not living on reserve
 - Civilians (non-military)

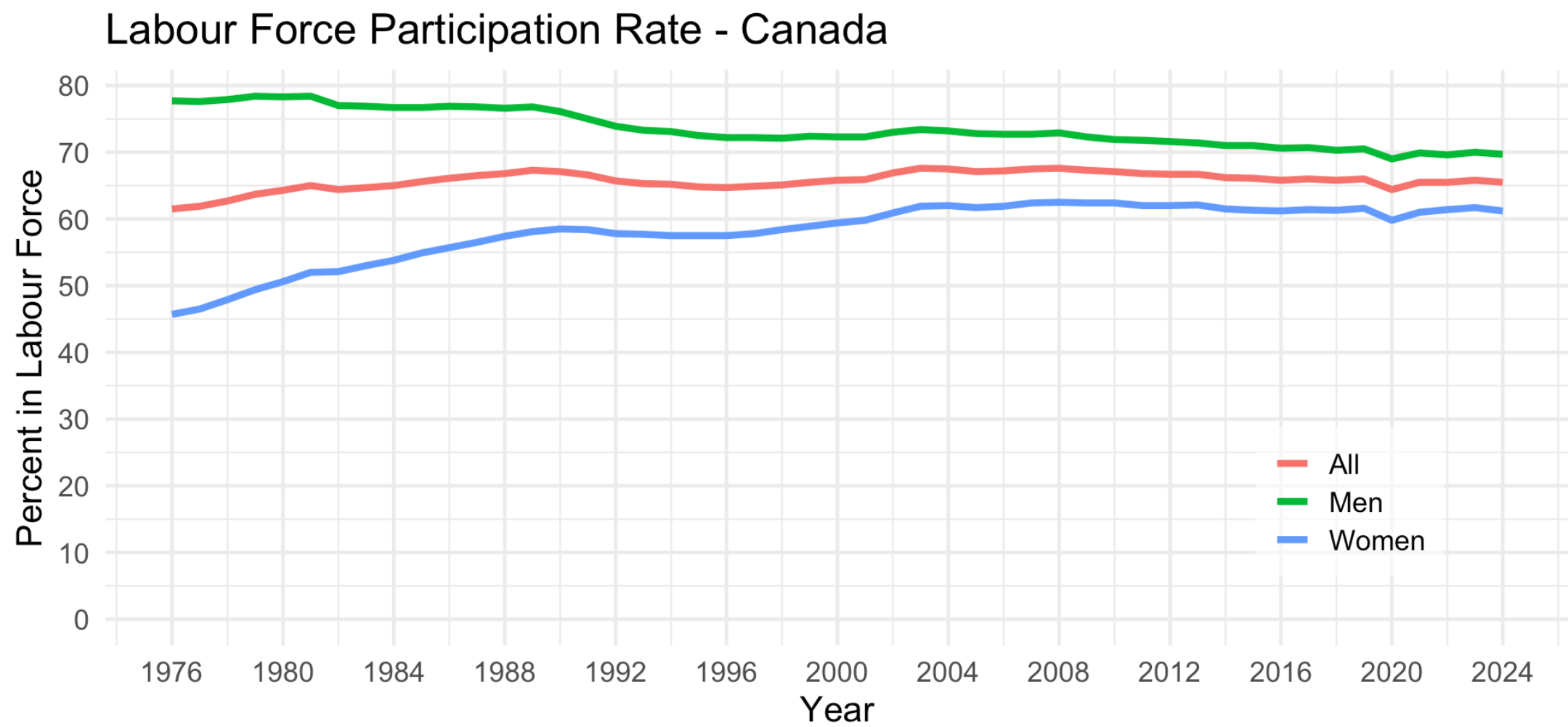
Labour Force Participation

- Eligible people who are not part of the labour force include
 - Retired people
 - People who do only unpaid work in the home
 - Full time students (who are 15 years old and over)
 - People who just give up on looking for work
- **Labour Force Participation Rate (LFPR):** The share of the eligible population (POP) in the labour force (LF)

$$LFPR = LF/POP$$

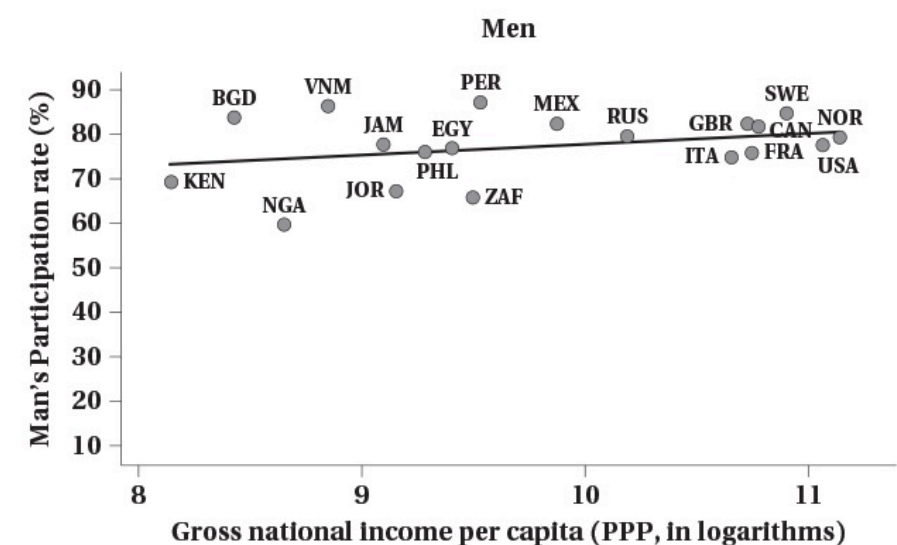
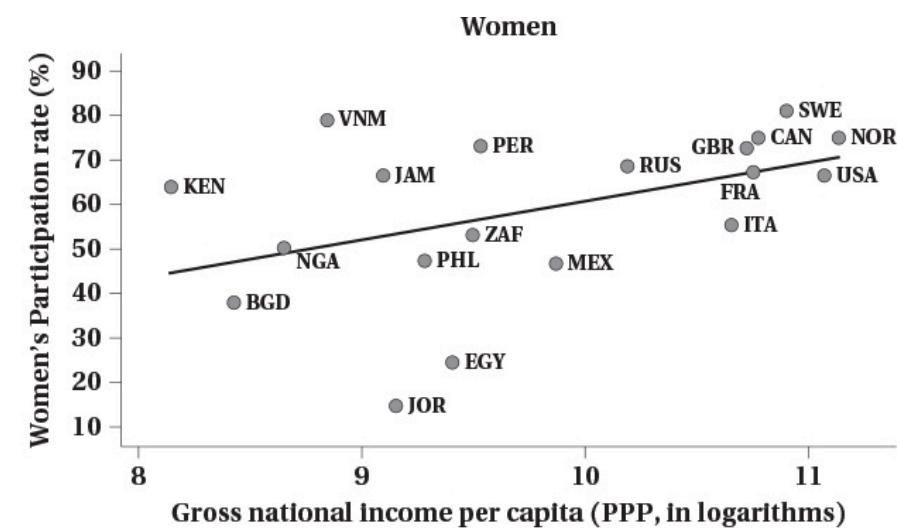
- The LFPR is a common statistic reported in the news

Labour Force Participation



Source: CANSIM Table 14-01-0327-02

Labour Force Participation



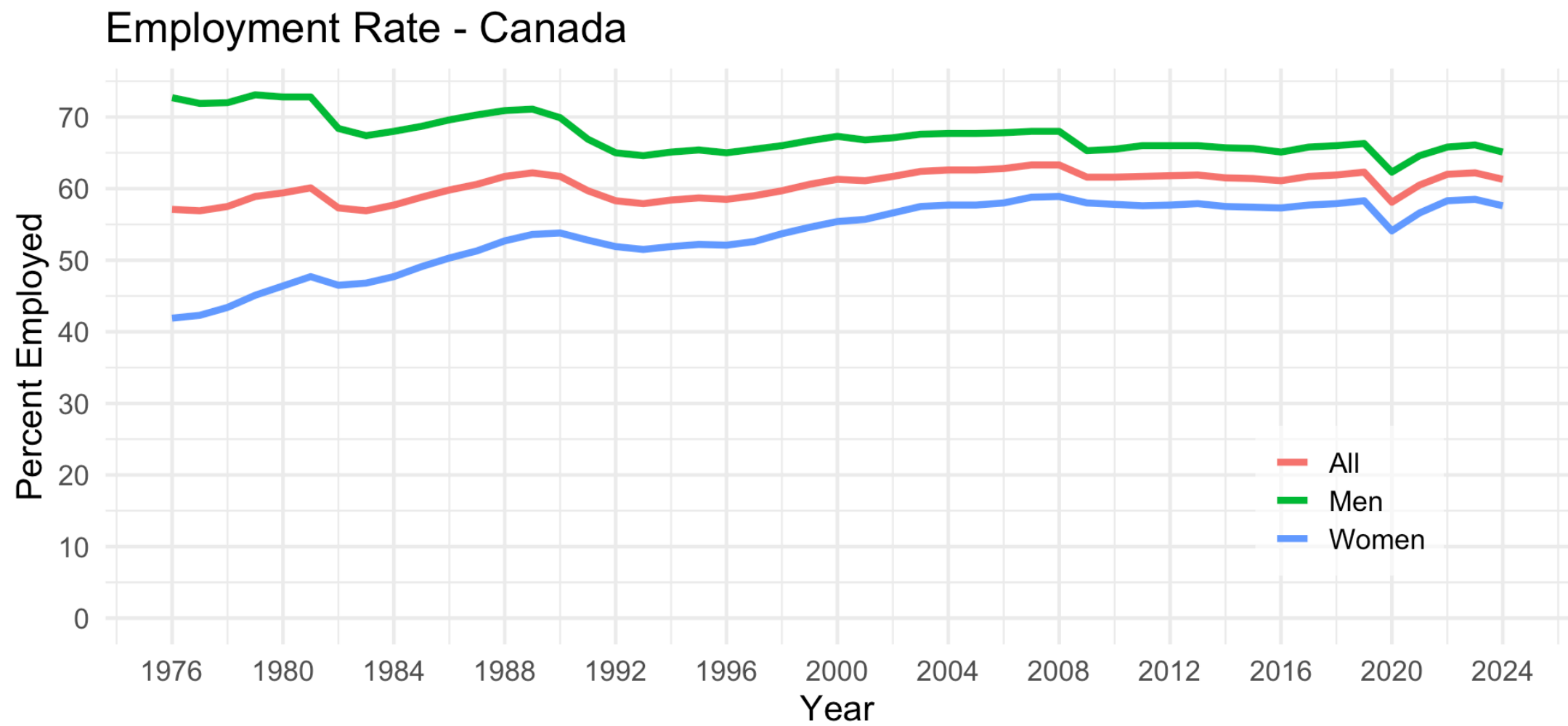
Employment and Unemployment

- People in the labour force are either employed or unemployed
- **Employed (E)**: people who worked during the reference period
 - Includes people normally employed but temporarily absent
- **Unemployed (U)**: people who did not work during the reference period but
 - were available for work and actively looked for work in the past four weeks
 - were waiting to start a new job within the next four weeks
 - on temporary layoff and expecting to be recalled to that job
- The employment rate (ER) and unemployment rate (UR) are

$$ER = E/POP$$

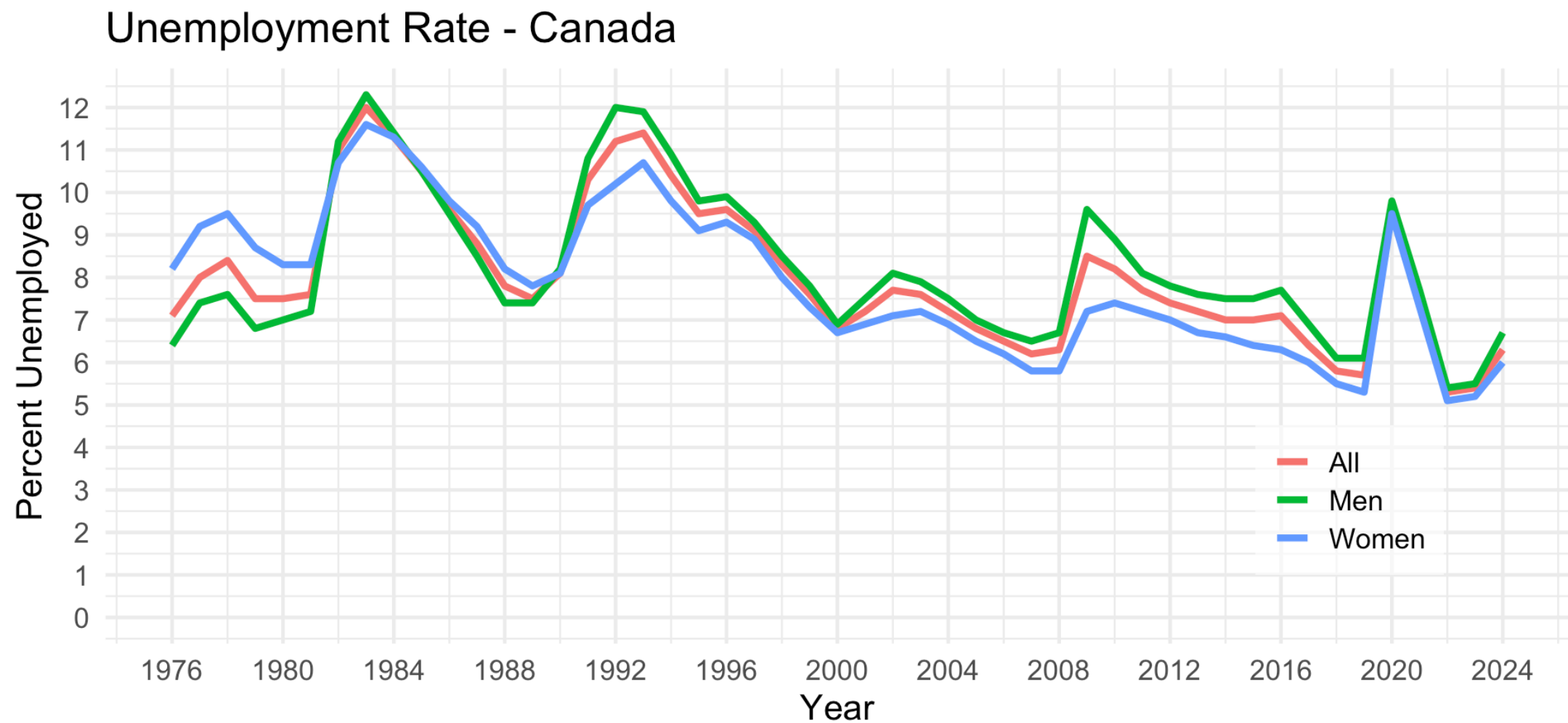
$$UR = U/LF$$

Employment and Unemployment



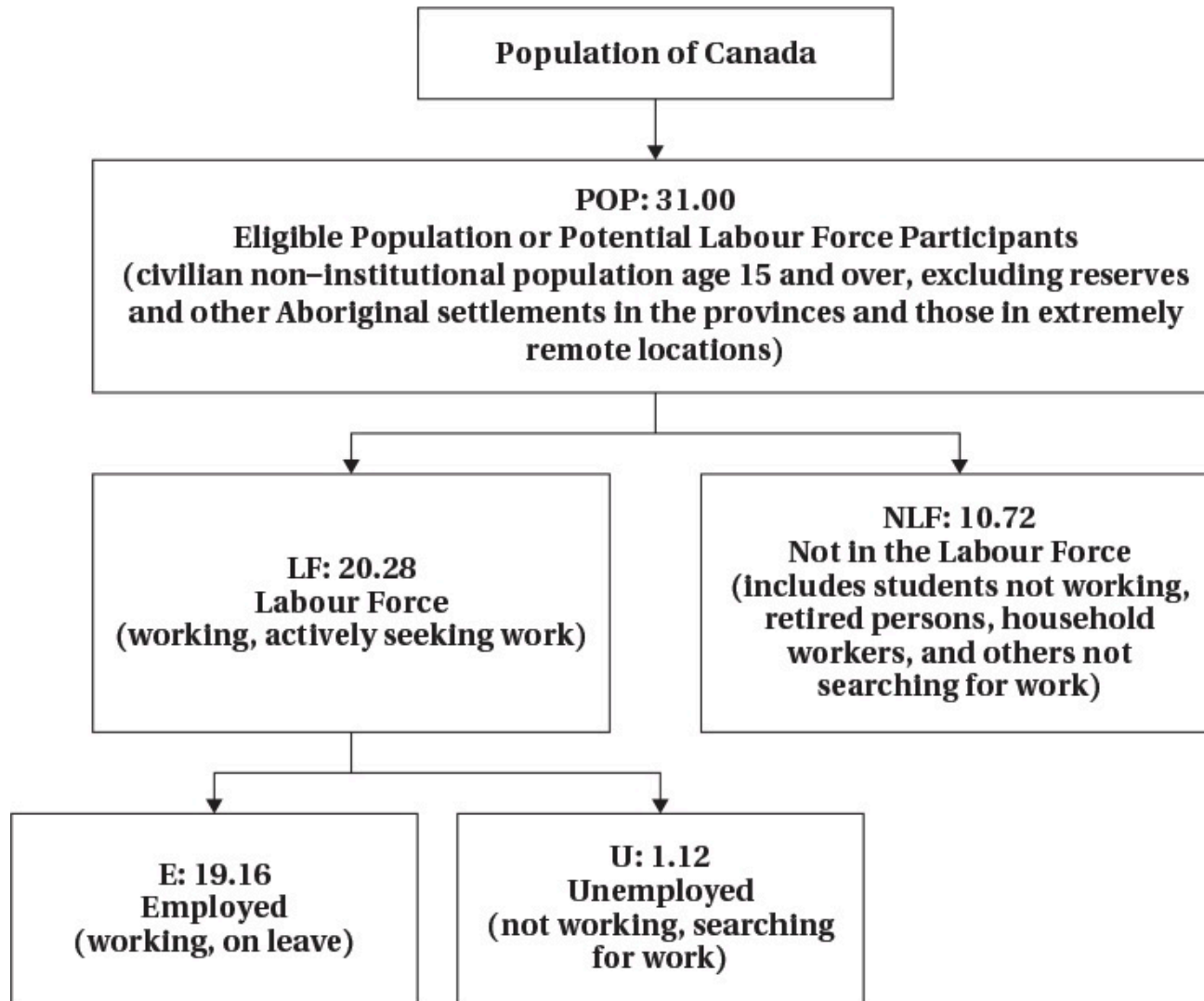
Source: CANSIM Table 14-01-0327-02

Employment and Unemployment



Source: CANSIM Table 14-01-0327-02

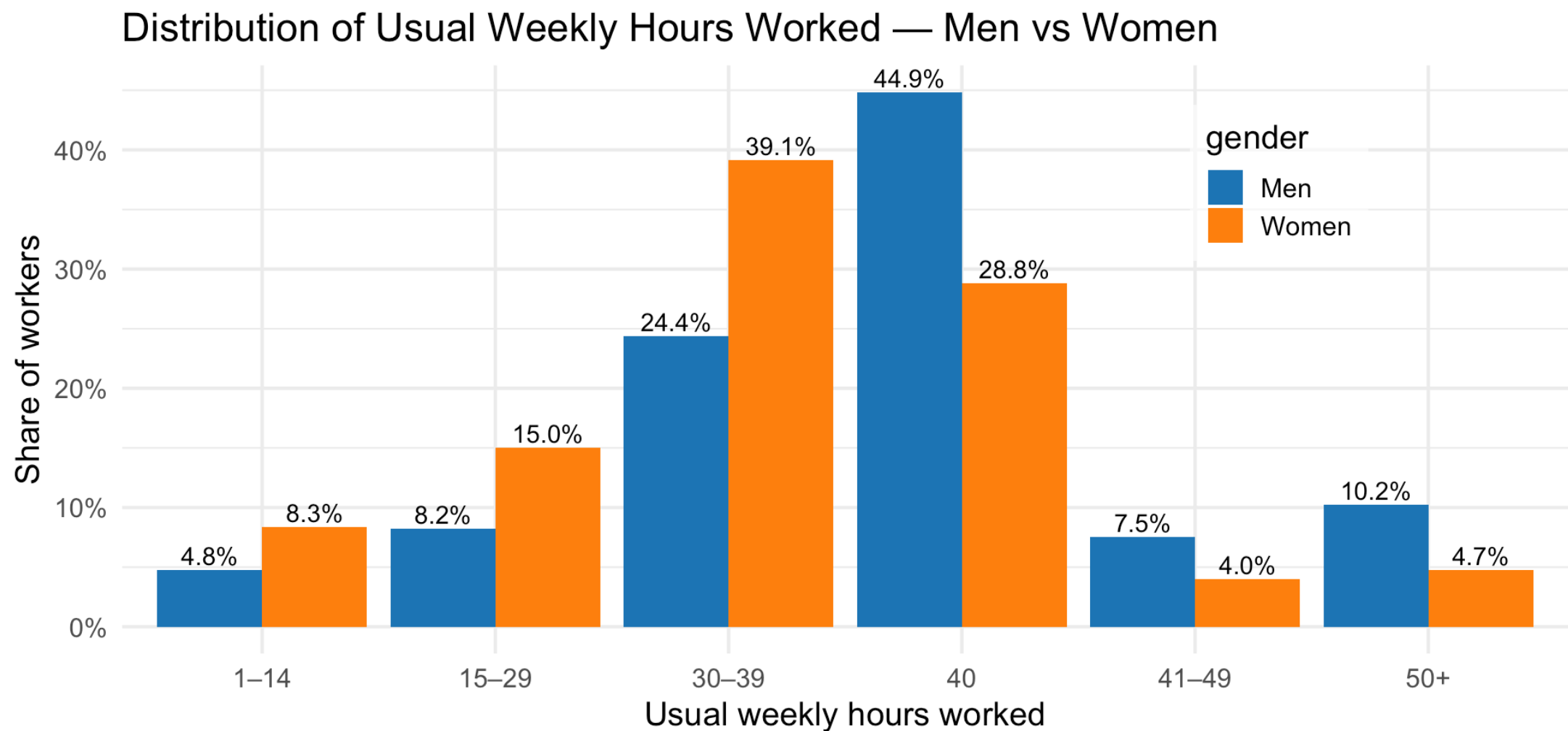
Labour Force Hierarchy



Hours Worked

- Individuals decide not only whether to work or not, but how intensively
- Some people work full time, some work part time
 - Work arrangements can be flexible
 - “Gig” jobs and remote work have increased that flexibility
- Recent data show
 - Almost half of men work full time hours
 - About 1/4 of women work full time hours

Hours Worked



Source: Labour Force Survey April 2025

Labour Supply Model

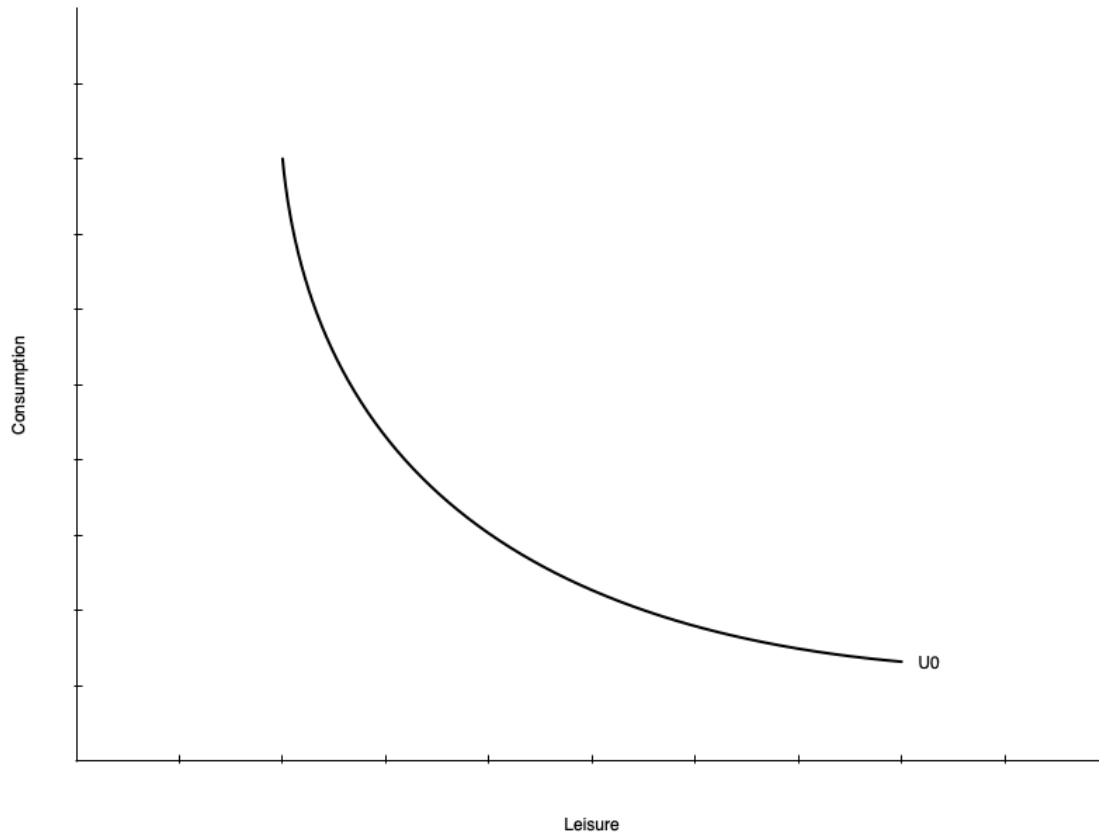
Introduction

- In this section we introduce a model we use throughout the course
- The model helps us understand how individuals make decisions about work
- Based on utility maximization as you saw in EC270
- Individual choose between two goods
 - Consumption of goods and services
 - Leisure
- Choices are constrained by time, wage rate, and non-labour income
- There are a few wrinkles that make this market unique

Preferences

- Individuals have preferences over consumption (C) and leisure (L)
 - People like consuming goods and services
 - Also like having non-work time
 - Includes fun activities but also non-work tasks like chores, education, etc.
- The tradeoffs people are willing to make between C and L are represented by indifference curves
 - At every point along the curve the individual is equally happy
 - Curves further from the origin represent higher levels of utility

Preferences

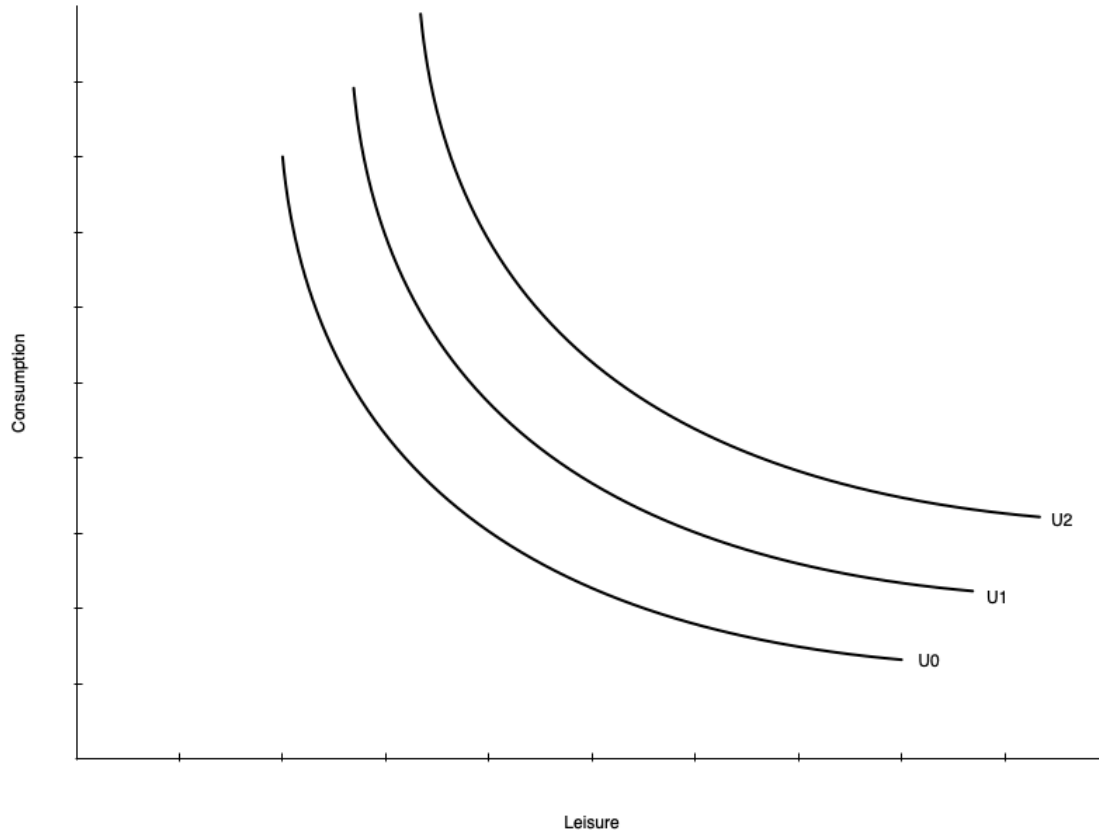


- On the left is a single indifference curve
- Any point represents equal utility (U_0)
- **Marginal Rate of Substitution (MRS):** the rate at which a person is willing to trade consumption for leisure while remaining equally happy
- The MRS is the slope of the indifference curve at a point
- Individuals have diminishing MRS
 - Willing to give up less consumption for additional leisure the more leisure they already have
 - Slope gets shallower with more leisure

Preferences

- Indifference curves do not have to be smooth curves like the one shown
- Could be straight lines
 - Means the tradeoff between consumption and leisure is constant
 - Willingness to trade away leisure for consumption does not depend on leisure
- Could be “L” shaped
 - Consumption and leisure must be consumed in fixed proportions
- Mostly we will deal with smooth preferences

Preferences

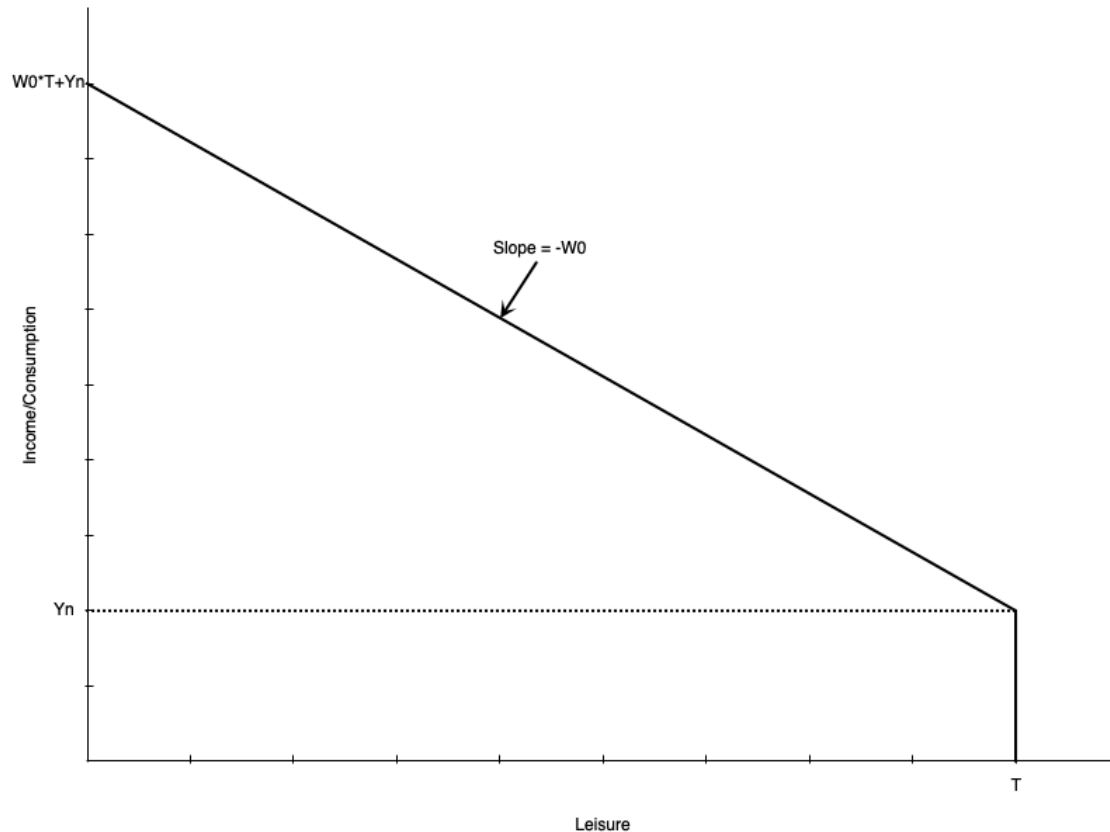


- Utility increases with curves further from the origin
 - $U_2 > U_1 > U_0$

Constraints

- Individuals do not choose consumption and leisure bundles on preference alone
- They are constrained by their budget
- The budget is determined by
 - The wage rate
 - Total number of hours available
 - Non-labour income
- Budget constraint is also called
 - Potential income constraint
 - Full income constraint

Constraints



- Graph shows a linear budget constraint
- Assume no saving, so consumption = income
- There are T total hours to allocate to work or leisure
- Individual earns Y_n non-labour income
 - Earned even with zero work hours (full leisure hours)
- Each hour worked gets wage of w
- Person working T hours (no leisure) earns $WT + Y_n$ income
 - Called “full income”

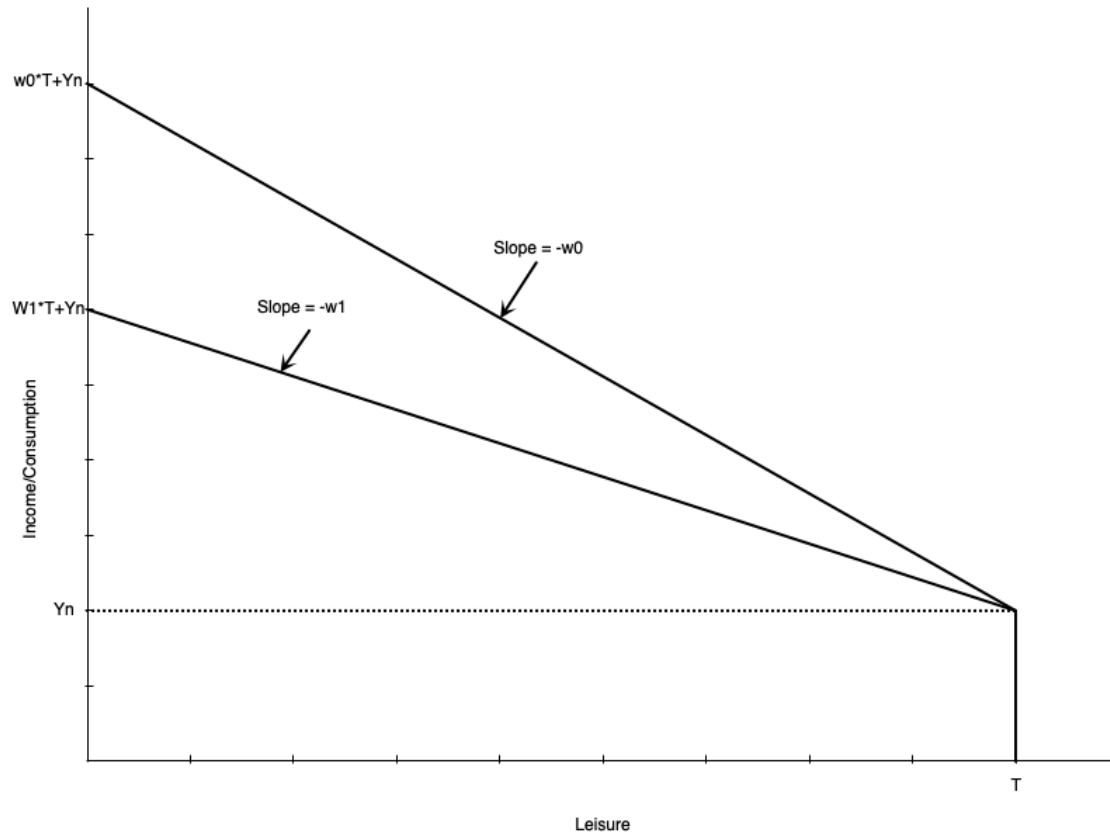
Constraints

- Mathematically, the budget constraint is

$$C = w(T - L) + Y_n$$

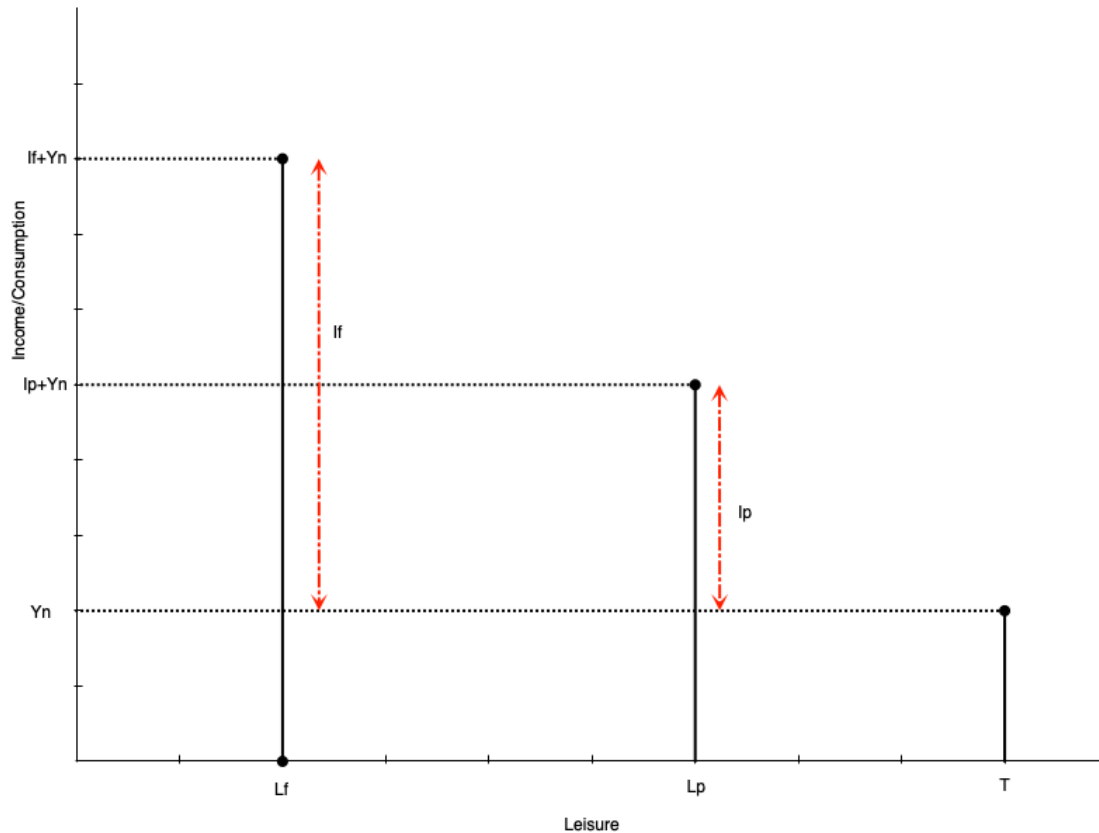
- Consumption is equal to income from working ($w(T - L)$) plus non-labour income (Y_n)
- The slope of the budget constraint is $-w$
 - The opportunity cost of leisure is the wage rate
 - Each hour of leisure means one less hour worked, which means w less income
- Someone who works zero hours has T leisure hours and Y_n consumption
- Someone who works T hours has zero leisure hours and $wT + Y_n$ consumption

Constraints



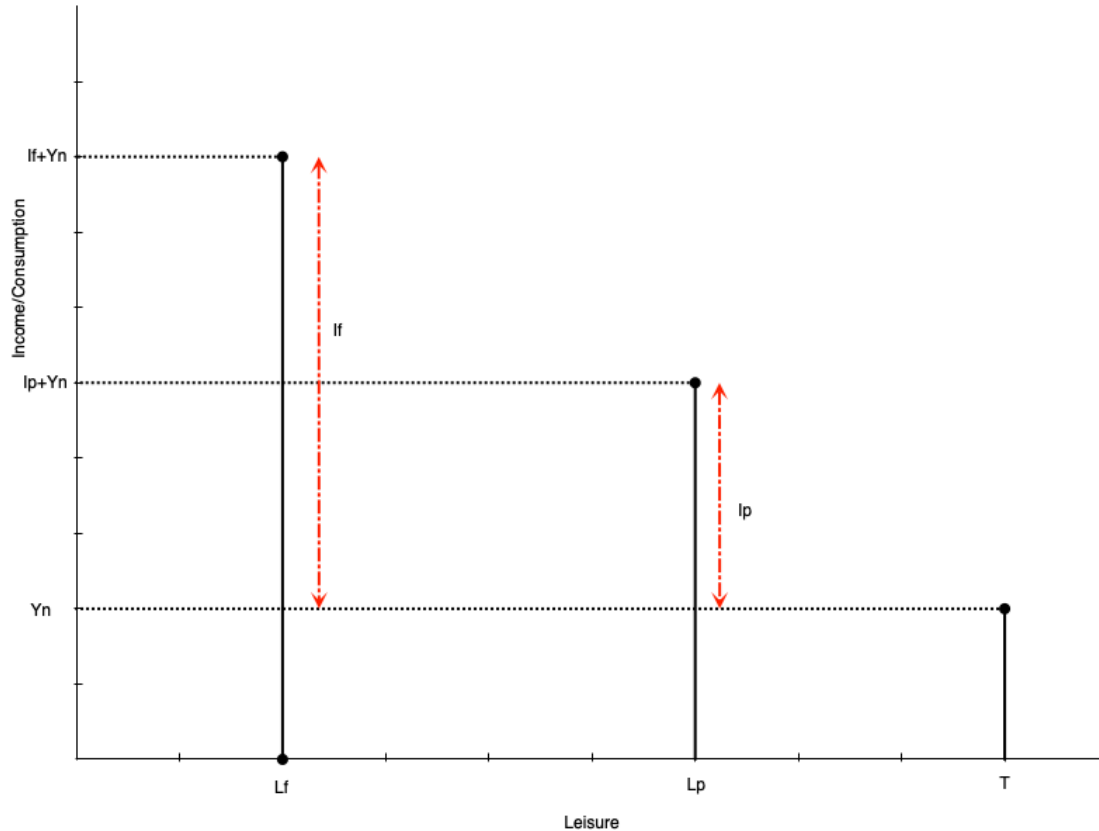
- Slope is the negative of the wage rate w
- A lower wage means shallower slope
 - If $w_1 < w_0$, the budget line swivels down
 - They intersect at leisure = T because there are no work hours
- Lower wage also means full income is lower
 - Working all hours leads to less total labour earnings

Constraints



- Often people cannot adjust their work hours freely
- They may only be able to work full time or part time
- Budget constraint is no longer a line, but a set of points
- Suppose an individual chooses part time work
 - Work hours are $h_p = T - L_p$
 - Labour earnings are $I_p = w * h_p$
 - Total consumption (and income) is $C_p = I_p + Y_n$

Constraints

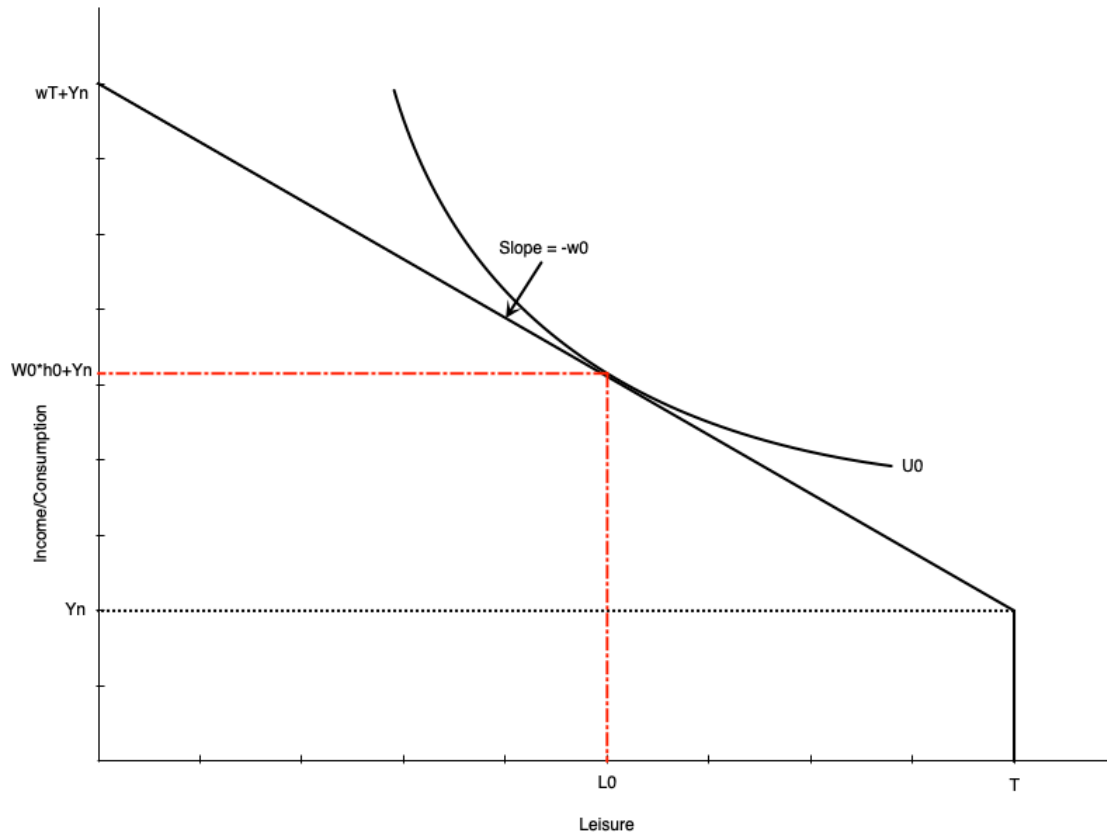


- A person working full time hours
 - Work hours are $h_f = T - L_f$
 - Labour earnings are $I_f = w * h_f$
 - Total consumption (and income) is $C_f = I_f + Y_n$
- In theory wage could also be different for full time work

Consumer Optimum

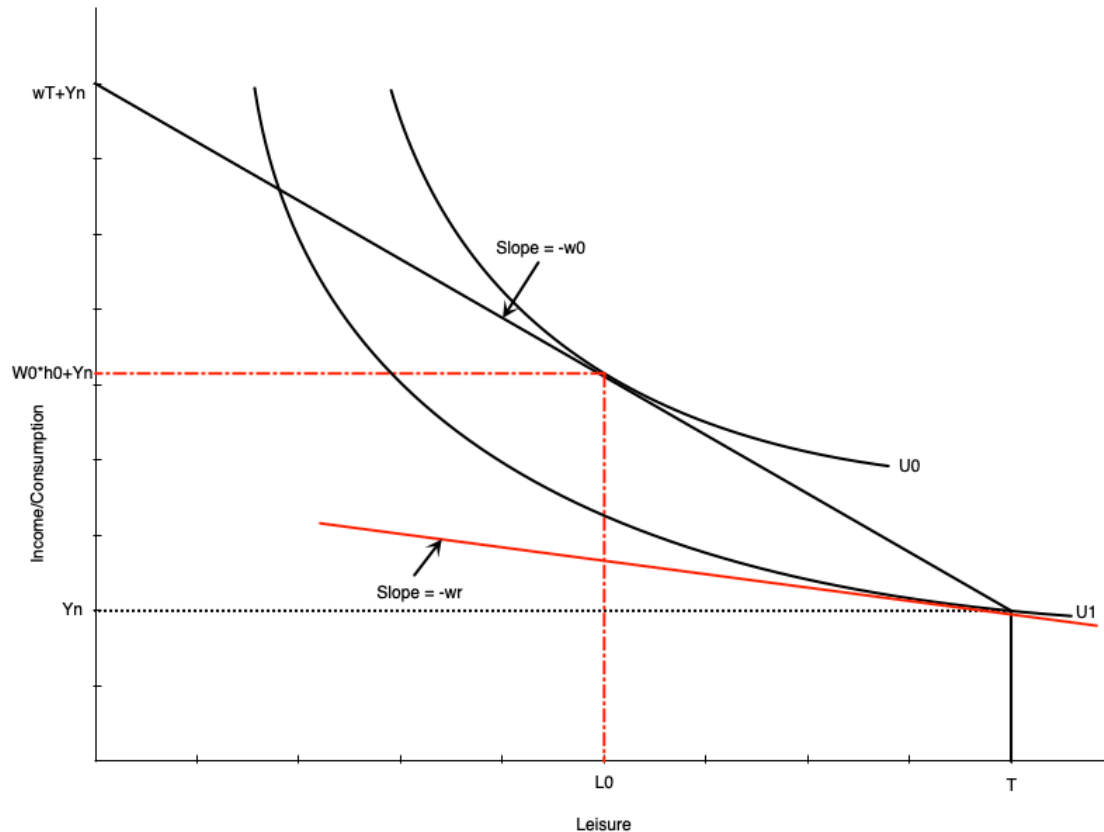
- Work hours are chosen using the combination of preferences and constraints
- The consumer optimum is the point that maximizes utility subject to the budget constraint
- The optimum occurs where the budget constraint is tangent to an indifference curve
- There are two possibilities for the optimum
 - Non-participation: optimum occurs at the end of the budget constraint (full leisure)
 - Participation: optimum occurs at an interior point (some leisure, some work)
- Model also defines the **reservation wage**
 - Lowest wage rate at which a person will choose to work

Consumer Optimum



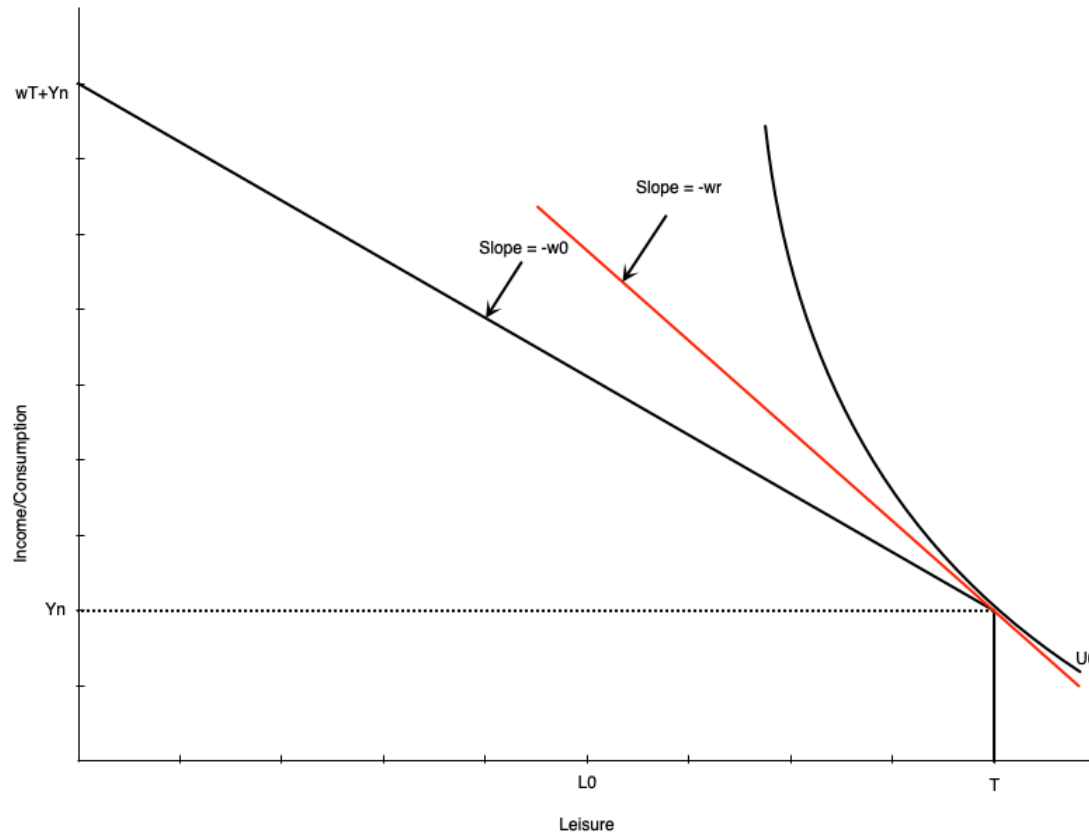
- Graph shows an optimum with participation
- Optimum is where indifference curve is tangent to budget constraint
- At the optimum, $MRS = \text{wage rate}$
 - Willingness to trade leisure for consumption (MRS) equals ability to trade them (wage)
- Optimal work hours are $h_0 = T - L_0$
- Optimal consumption is $C_0 = W_0 h_0 + Y_n$

Consumer Optimum



- The reservation wage is the lowest wage at which a person will choose to work
- It occurs where the budget constraint is tangent to the indifference curve at full leisure
 - At that point, $MRS = \text{reservation wage}$
- If the wage is any lower, the person will choose not to work
- If it is any higher, they will work positive hours
- The red line shows the hypothetical reservation wage
 - In this case, it is w_r

Consumer Optimum



- Graph to the left shows a non-participation optimum
- The slope of the indifference curve is steeper than the budget constraint at full leisure
 - The extra consumption you want from one less hour of leisure (MRS) is more than you can get from working an hour (wage)
- So your optimal choice is to not work at all
- Indifference curve touches the budget constraint at the kink
- Notice the wage w_0 is lower than the reservation wage w_r

Comparative Statics

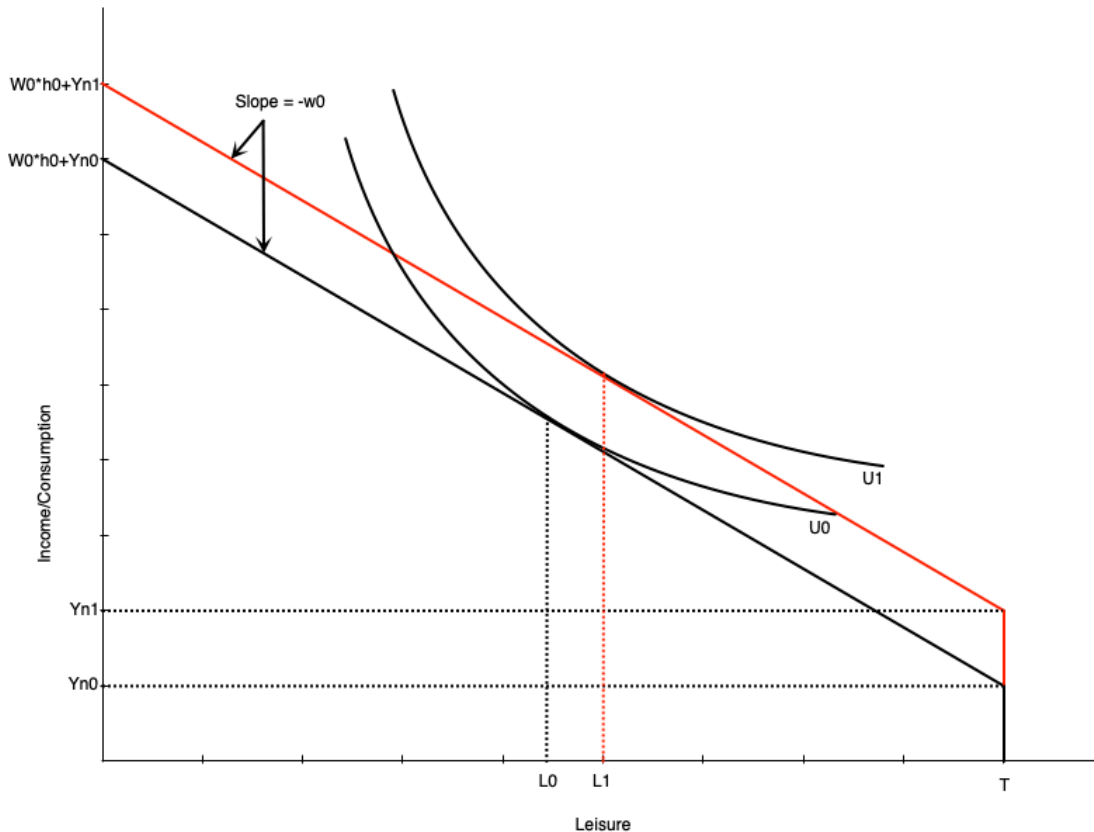
Introduction

- The model can be used to predict how changes in the environment affect work decisions
- Changes could be to the wage or non-labour income
- Often these arise because of policy changes
 - Introduction or increase in welfare payment
 - Wage subsidy
 - Tax credits
 - Employment insurance
 - Child care
- We can examine these change with our model

Increase in Non-Labour Income

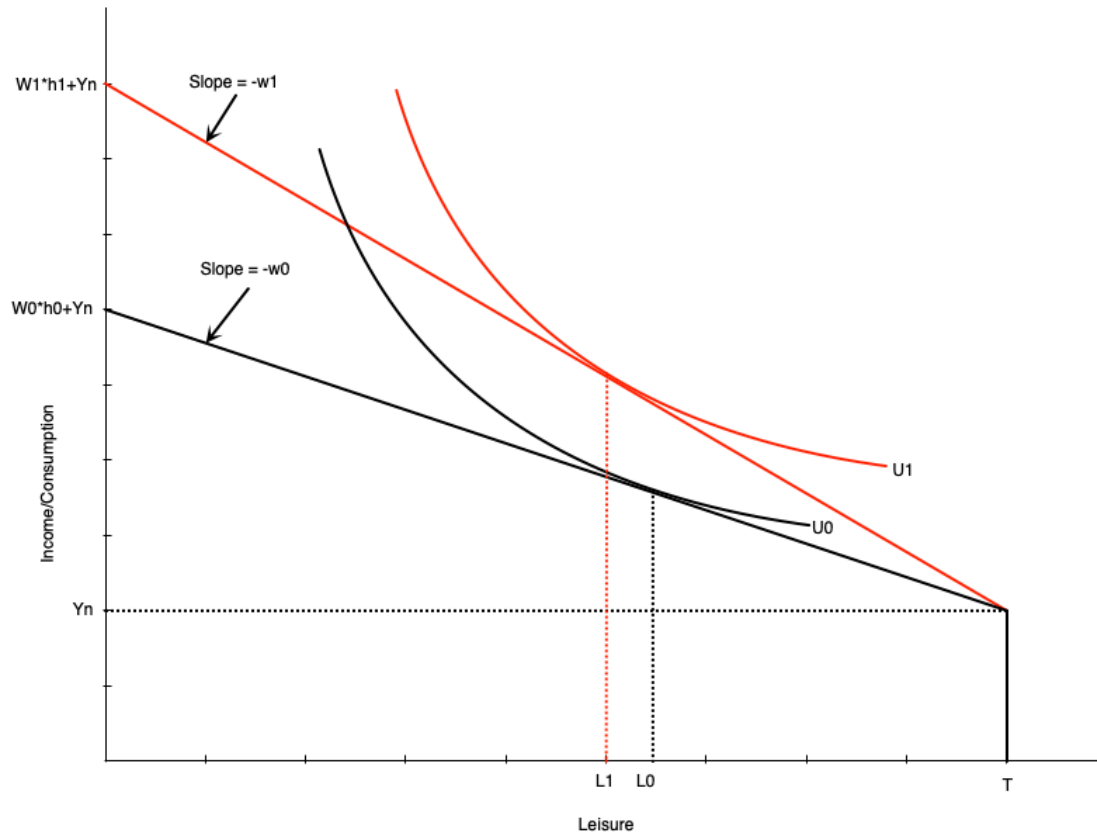
- Suppose that individuals receive an increase in non-labour income
 - Could be from a government transfer program
- How does this affect optimal work hours?
- Depends on whether leisure is a normal or inferior good
 - Normal good: leisure increases when non-labour income increases
 - Inferior good: leisure decreases when non-labour income increases
- Generally economists think of leisure as normal

Increase in Non-Labour Income



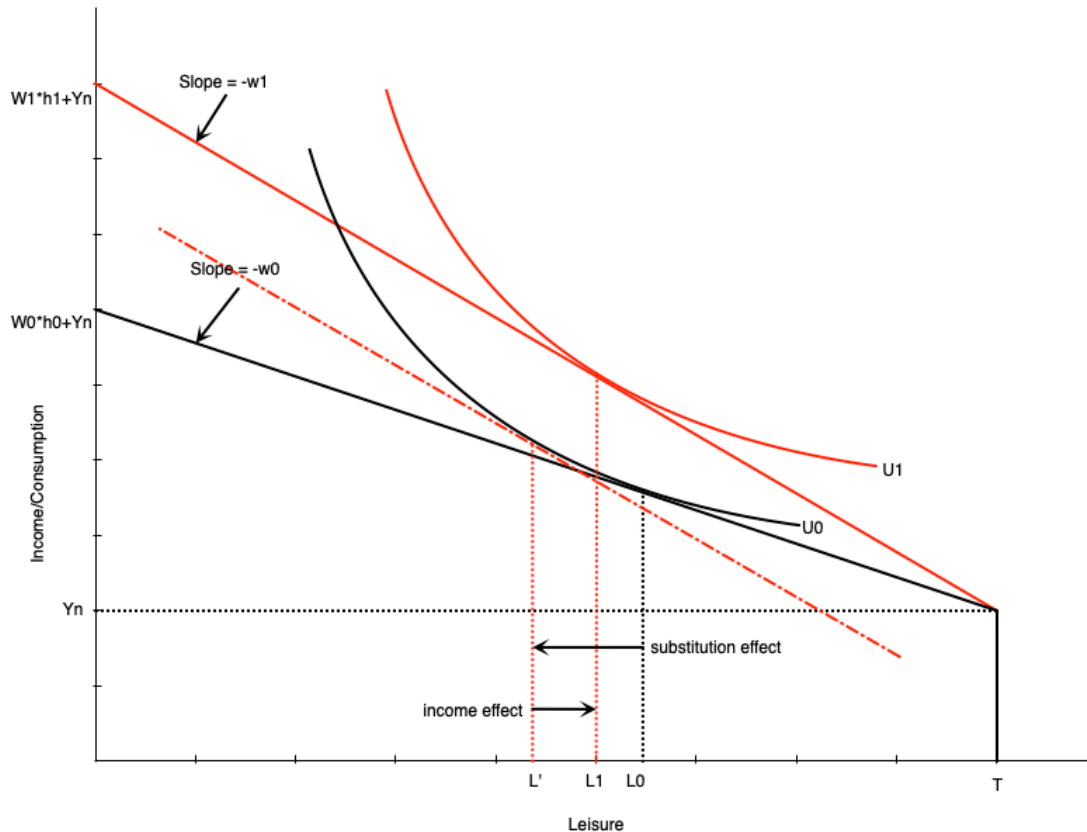
- Graph shows an increase in non-labour income from Y_{n0} to Y_{n1}
- Budget line shifts up vertically but parallel
 - No change in slope because wage is the same
- If leisure is a normal good, optimal leisure increases from L_0 to L_1
- Means work hours decrease by the same amount
- Consumption also increases
- This illustrates a pure **income effect**
 - Increases in income increase demand for normal goods

Increase in the Wage



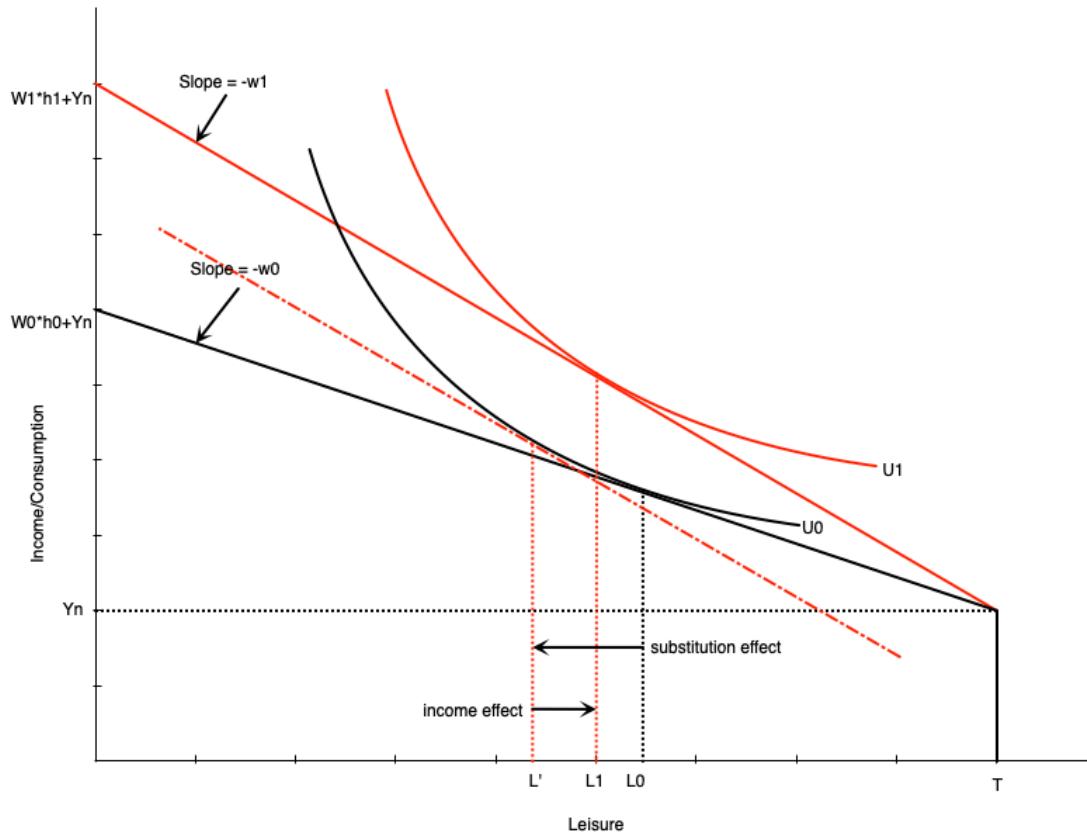
- If the wage changes, the budget constraint pivots
- In graph to the left, the wage increases from w_0 to w_1
- The slope becomes steeper
- Full income also increases because working all hours earns more
- Graph is drawn such that leisure falls from L_0 to L_1 at the new wage rate
- There are two effects underlying this change
 - Substitution effect
 - Income effect

Increase in the Wage



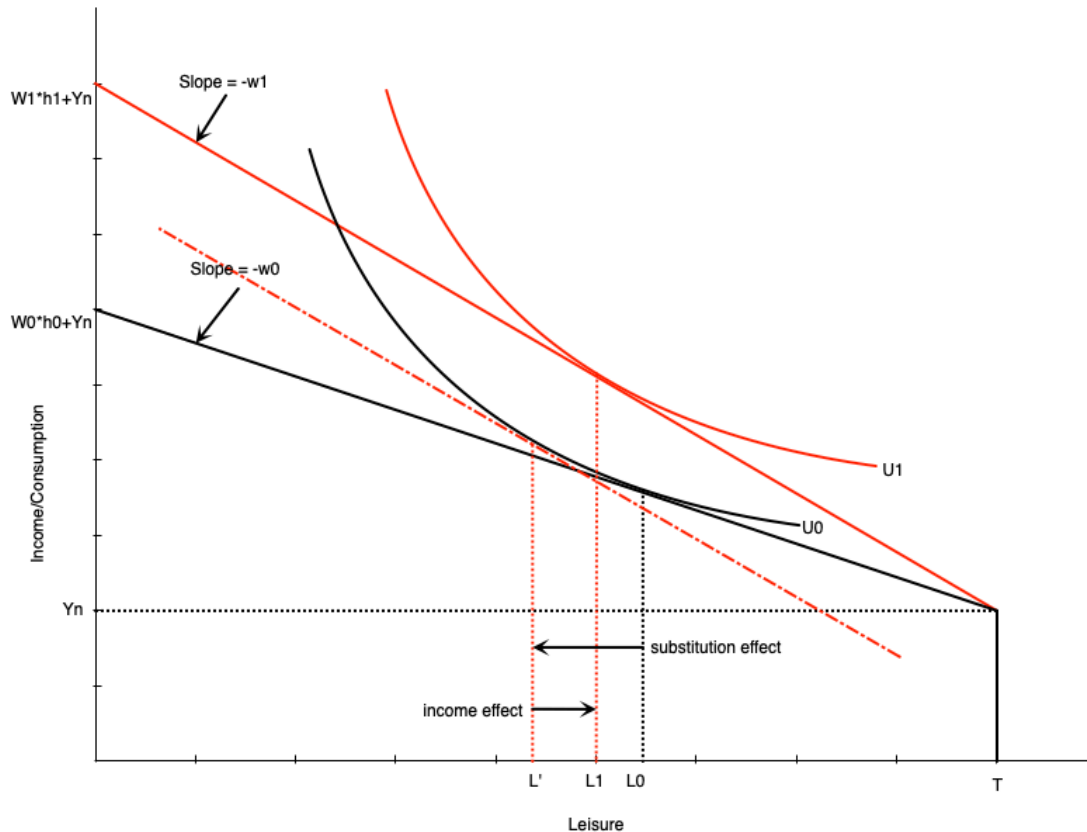
- **Substitution Effect:** change in leisure when wage changes but utility is held constant
 - When wage increases, leisure becomes more expensive
 - People substitute away from leisure and towards consumption
 - To measure this, draw a hypothetical budget line with the new slope but tangent to the original indifference curve
 - Difference in leisure measures substitution effect

Increase in the Wage



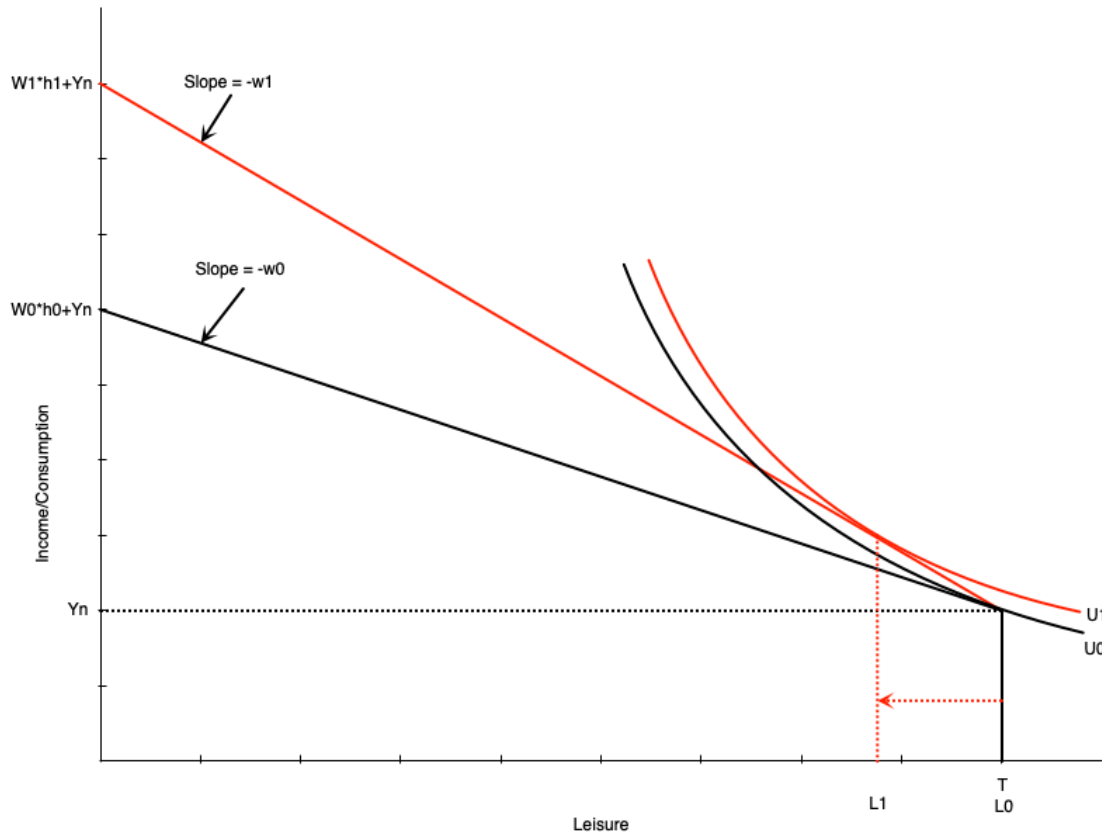
- **Income Effect:** change in leisure when income changes but wage is held constant
 - When wage increases, workers become richer at same hours of work
 - Want to consume more of all normal goods
 - To measure this, move from hypothetical budget line to new budget line
 - Difference in leisure measures income effect

Increase in the Wage



- The income and substitution effects move in opposite directions
- For a wage increase
 - Substitution effect encourages more work (less leisure)
 - Income effect encourages less work (more leisure)
- The overall effect on work hours depends on which effect is stronger
- Graph on the left shows a stronger substitution effect

Increase in the Wage

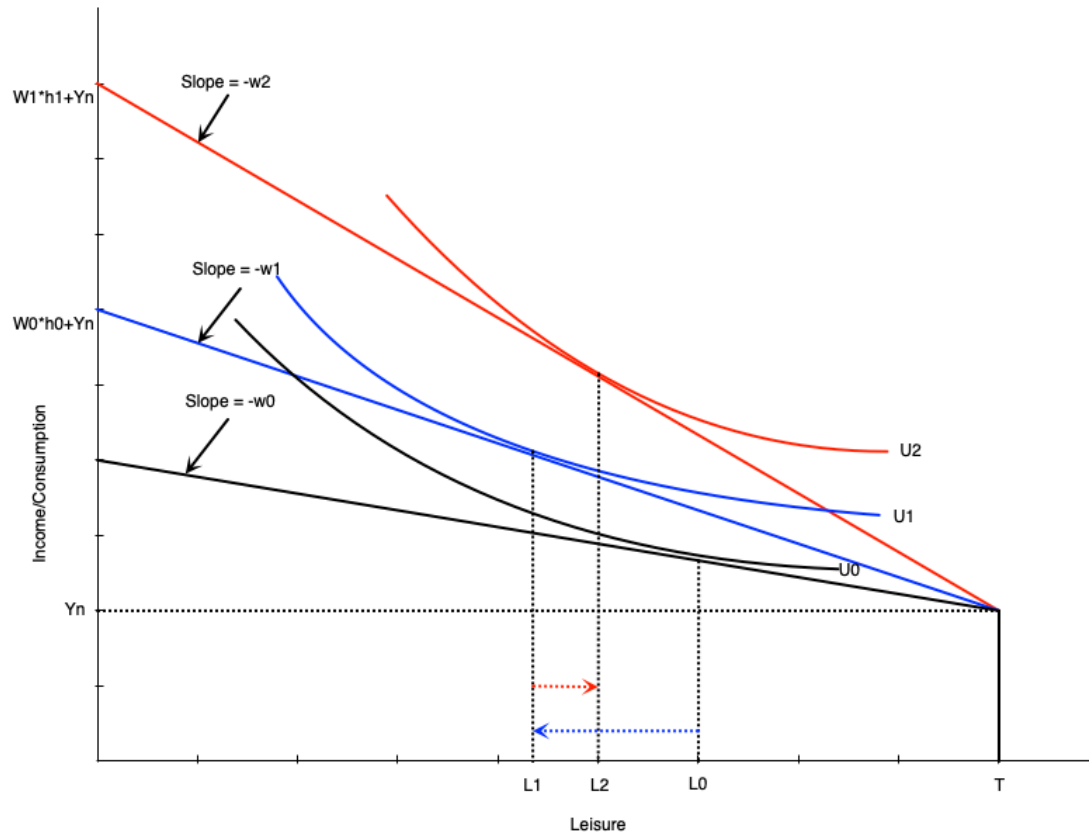


- Changing the wage can also affect participation
- Recall that workers do not work if the wage is below their reservation wage
- If an increase in the wage moves it above the reservation wage, they will start working
- Graph to the right shows how this would happen

The Individual Labour Supply Curve

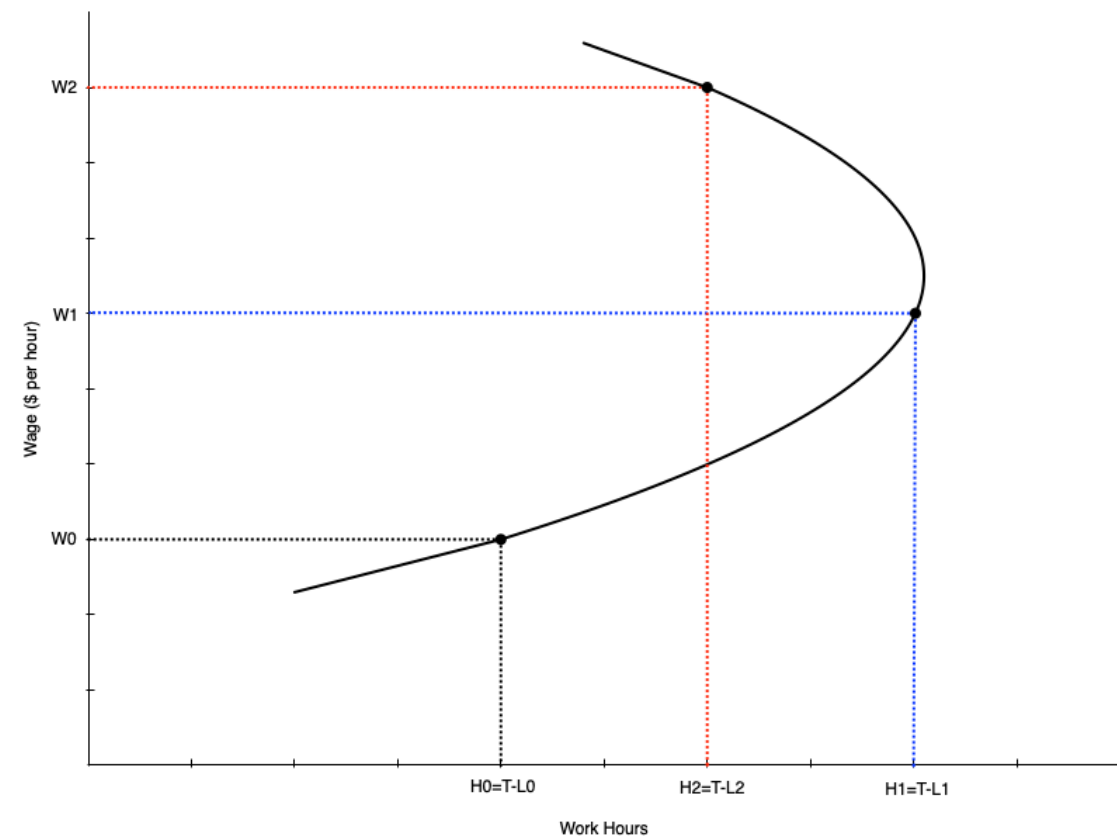
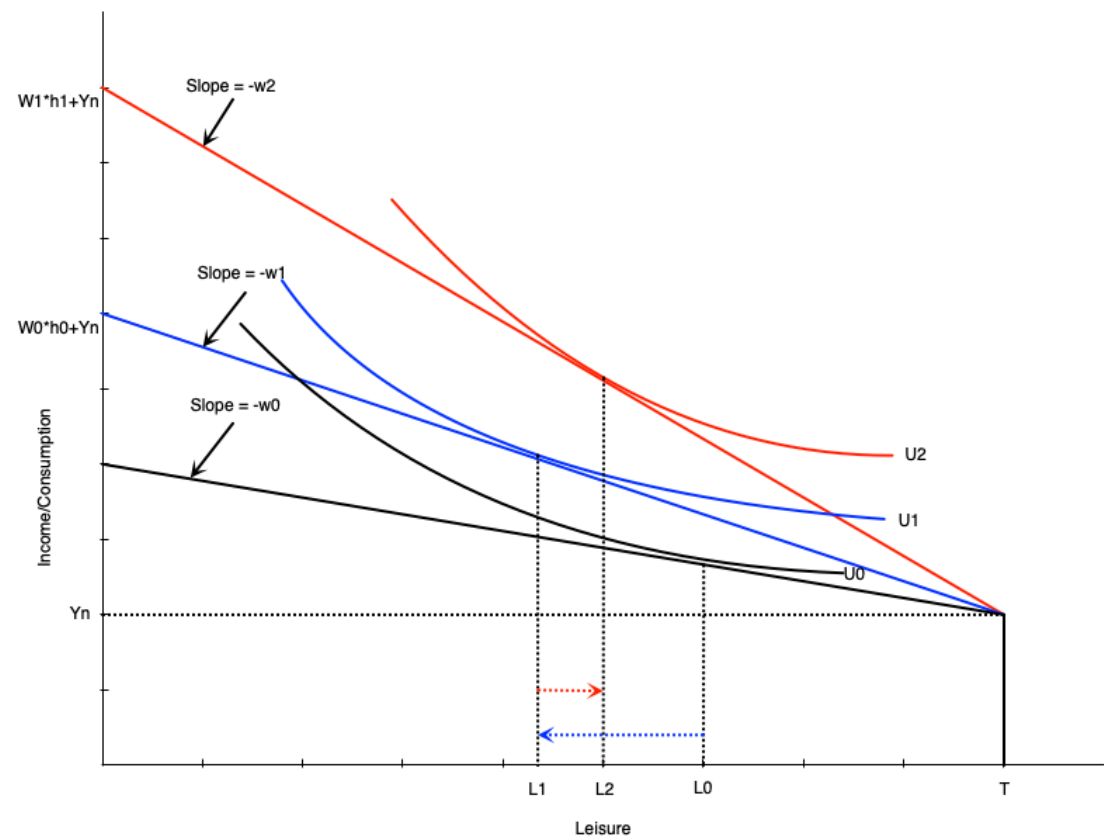
- The labour supply curve shows the relationship between the wage rate and hours worked
- It is derived from the consumer optimum at different wage rates
- We will see that the labour supply curve can be backward bending
 - It is upward sloping when the substitution effect is stronger
 - It is downward sloping when the income effect is stronger
 - The curve tends to bend backwards at higher wage rates

The Individual Labour Supply Curve



- Suppose wage goes from w_0 to w_1 to w_2
- At w_0 , work hours are $h_0 = T - L_0$
- At w_1 , work hours increase to $h_1 = T - L_1$
 - substitution effect stronger
- At w_2 , work hours decrease to $h_2 = T - L_2$
 - income effect stronger

The Individual Labour Supply Curve



Summary

- We have used the labour supply model to examine changes in the wage
- For people participating, wage changes alter leisure and work hours
- If the wage goes up then
 - Work hours go up if the substitution effect is strongest
 - Work hours go down if the income effect is strongest
- For non-participants, a wage increase can induce them to enter the labour force
 - If the wage goes above their reservation wage
- The labour supply curve maps different wages to work hours
- It slopes up at lower wages, but can bend backwards at higher wages

Empirical Evidence

Introduction

- There are many labour economists working on evidence of these theories
- We will look at a few examples of empirical evidence on labour supply
- First we look at evidence on participation
 - The **extensive margin** of labour supply
 - Many studies look specifically at married women
- Then at evidence on work hours
 - The **intensive margin** of labour supply

Participation

- Table below shows participation rates for married women
- Broken down by age, children, education, spousal income
- Each of these demographic factors are related to wage
 - e.g. higher education -> higher wage
- We expect to see that where wage is higher, participation is also higher
- This is not a perfect way to examine this relationship
 - There could be other related factors that drive participation

Participation

TABLE 2.2 Labour Force Participation Rates of Married Women, Canada 2017

	Percentage of Married Women	Participation Rate ^a	Raw Difference from Base Group ^b	Adjusted Difference from Base Group ^c
All Women, Total		75.3		
<i>Age</i>				
20–24	1.1	89.4	N/A	N/A
25–34	18.9	84.4	–5.0	–3.7
35–44	23.9	85.5	–3.9	–3.0
45–54	24.0	84.8	–4.6	–7.3
55–64	23.8	64.8	–24.6	–22.3
65–69	8.4	27.5	–61.9	–52.6
<i>Age of youngest child, under 18 years old, at home</i>				
No children under 18	60.1	70.9	N/A	N/A
0–5 years old	18.4	77.4	6.5	–13.9
6–9 years old	7.7	84.6	13.7	–5.4
10–15 years old	10.1	87.0	16.1	–0.7
16–17 years old	3.7	86.3	15.4	0.2
<i>Education</i>				
Less than high school graduation	8.2	49.6	N/A	N/A
High school graduate or some post-secondary	23.6	67.9	18.3	12.1
Non-university, post-secondary certificate	30.7	77.8	28.2	18.8
University degree	37.6	83.7	34.1	22.4
<i>Spouse's income</i>				
None (or negative)	18.0	49.7	N/A	N/A
\$1–\$9,999	8.7	62.0	12.3	7.7
\$10,000–\$24,999	9.1	78.6	28.9	17.5
\$25,000–\$49,999	18.2	82.3	32.6	18.4
\$50,000–\$74,999	17.1	87.1	37.4	21.0
\$75,000–\$99,999	12.4	86.3	36.6	18.9
\$100,000–\$149,999	10.7	84.2	34.5	17.1
Over \$150,000	5.7	74.1	24.4	7.4

Elasticity of Labour Supply

- The **elasticity of labour supply** is the percentage change in labour supply from a 1% change in the wage
- Measures how responsive work hours are to changes in the wage
- There are different ways of measuring elasticity
 - **Uncompensated Elasticity:** measures total change in work hours from a wage change
 - Includes both income and substitution effects
 - Effect is theoretically ambiguous
 - **Compensated Elasticity:** measures change in work hours after removing income effect
 - Isolates the substitution effect
 - Should be positive

Elasticity of Labour Supply

TABLE 2.3 **Estimated Elasticities of Labour Supply**

Sex	Uncompensated (Gross, Total) Wage Elasticity of Supply	Compensated (Substitution) Wage Elasticity of Supply	Income Elasticity of Supply
Both sexes	0.0 to 0.2	0.1 to 0.3	−0.1 to 0.0
Men and single women	−0.1 to 0.2	0.1 to 0.3	−0.1 to 0.0
Married women	0.1 to 0.3	0.2 to 0.4	−0.1 to 0.0

NOTES: The uncompensated elasticity represents the elasticity for the choice of hours to work, conditional on working.
SOURCE: Adapted from Table 2 of McClelland and Mok (2012), in their review of U.S. studies for the Congressional Budget Office.

Elasticity of Labour Supply

- Table above shows summary of uncompensated and compensated elasticities from different studies
- Separates by gender
 - We might expect men and women to respond differently to wage changes
 - Women potentially have more flexibility in work hours and respond more
- Table shows uniformly *inelastic* labour supply
 - Elasticity less than 1 in absolute value
 - A 1% increase in wage leads to roughly 0.1% to 0.3% change in work hours
- Compensated elasticity is positive
 - Women respond more to wage changes
- Income effect is generally negative or small

Applications

Introduction

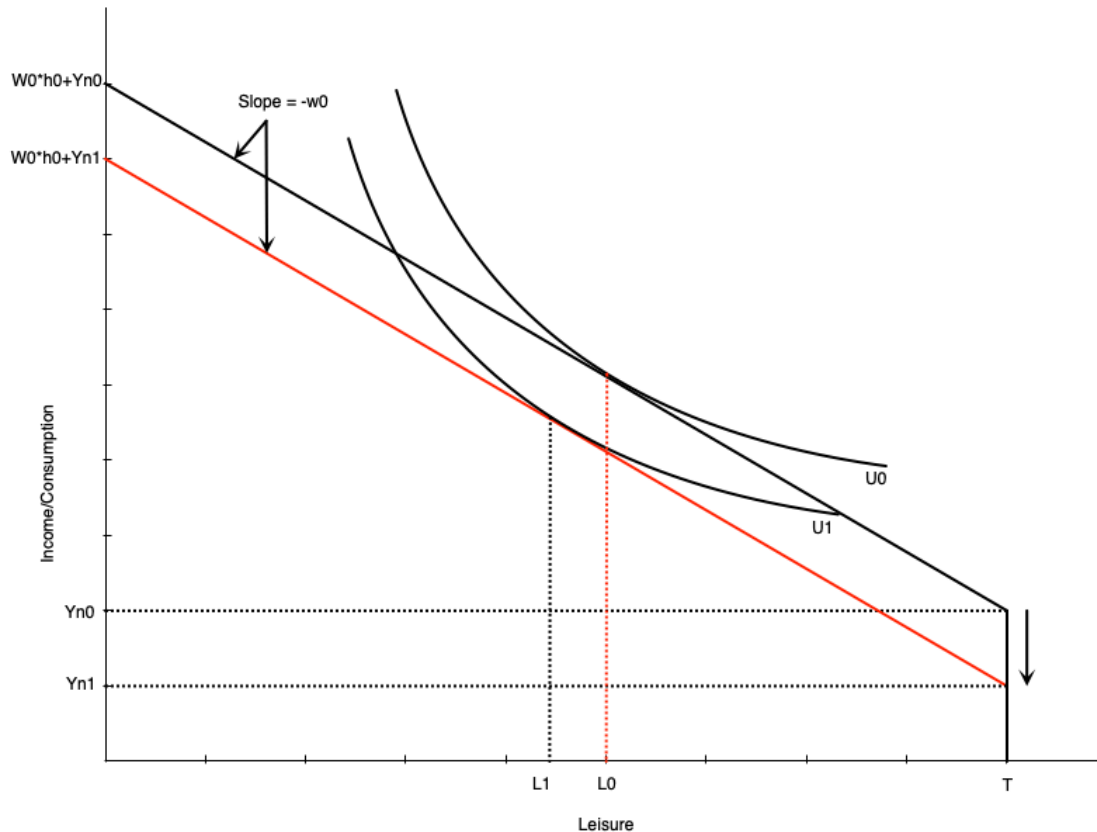
- We can use the model to analyze various situations faced by workers
- In this section we will look at
 - Changes in family income (added worker effects)
 - Discouraged workers
 - Moonlighting (holding multiple jobs)
 - Overtime
 - Flex work
- The list is not exhaustive
- In the next chapter we will look at government policies

Added Worker Effect

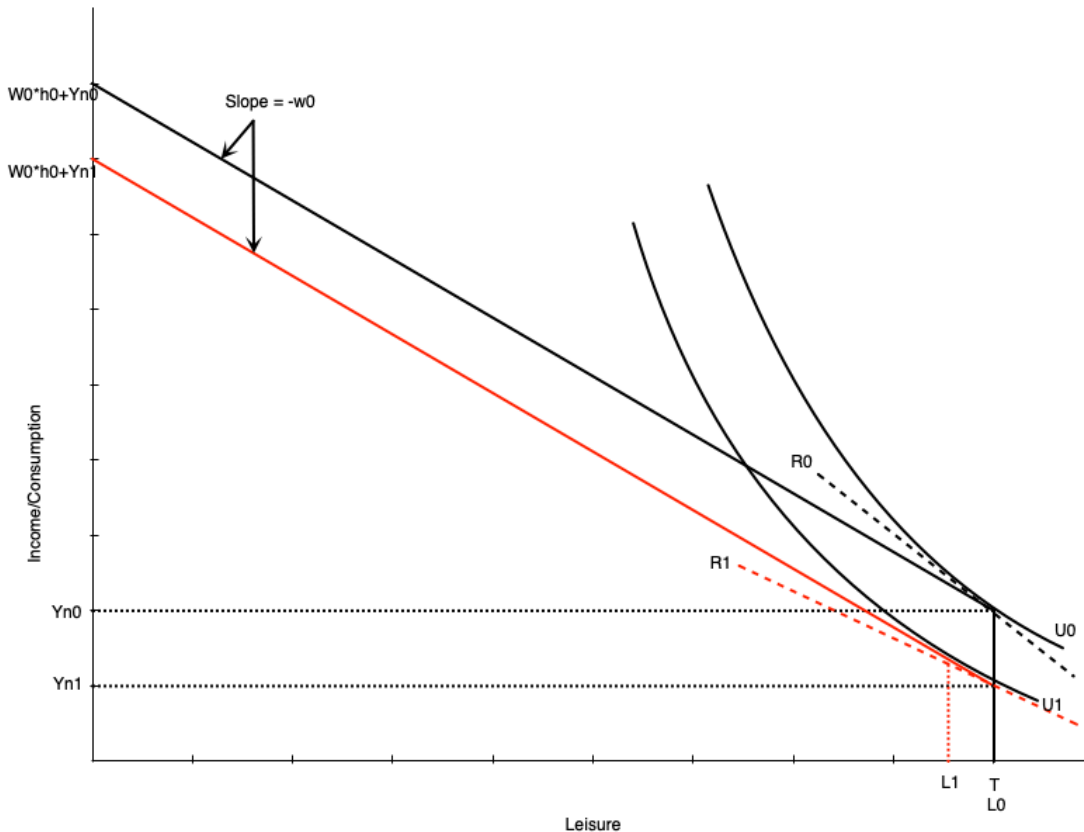
- The **added worker effect** occurs when a negative income shock to a family leads to increased labour supply from secondary earners
 - e.g. husband loses job, wife enters labour force or increases work hours
- Explicitly considers the family unit rather than treating labour supply as purely individual
- Spousal income is treated as a person's non-labour income
- In the model we can look at what happens when this amount falls

Added Worker Effect

- When non-labour income falls, the budget constraint shifts down
- If leisure is a normal good, optimal leisure falls from L_0 to L_1
- Work hours increase from h_0 to h_1
- Means that the spouse increases work hours to compensate for lost family income



Added Worker Effect

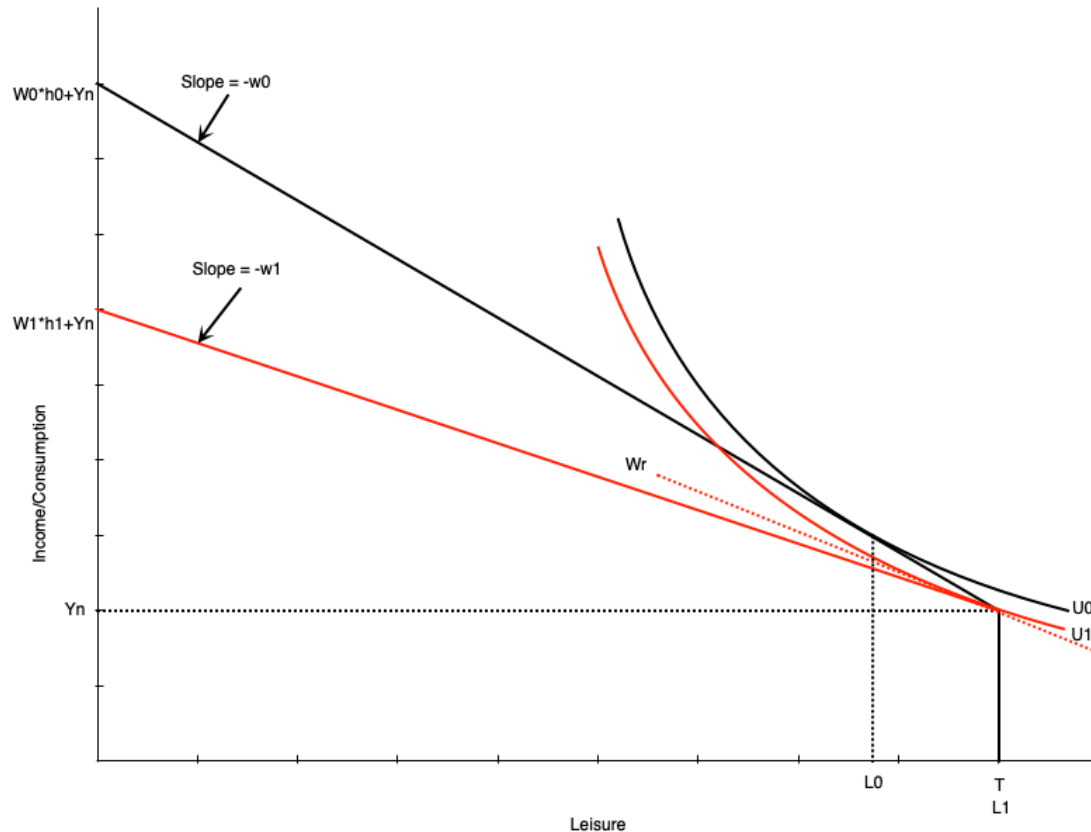


- If the spouse was not already working, the fall in income can induce participation
- Happens because the fall in income reduces the reservation wage
 - If the wage is now above the reservation wage, they will start working
- Graph to the left shows how this happens
 - The slope of reservation wage R_1 is smaller than R_0 after the drop in non-labor income
 - Person enters the labour market in response because wage is now above reservation wage

Hidden Unemployment

- We discussed a problem with the unemployment rate in that it does not include discouraged workers
- **Discouraged workers:** people who could work but have given up looking and dropped out of the labour force
- Our model can explain this behaviour
- In bad times, the wage falls
- If the wage falls below a person's reservation wage, they will drop out of the labour force

Hidden Unemployment

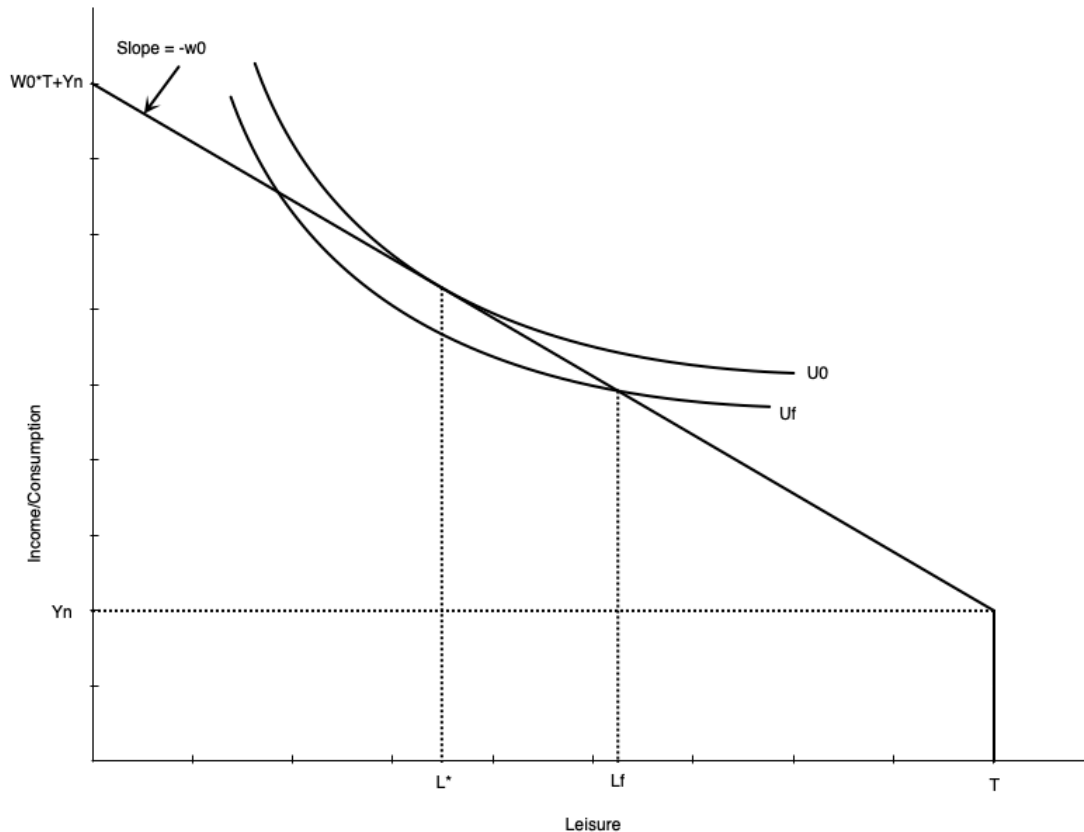


- Graph shows worker initially working positive hours
- Wage falls from w_0 to w_1
- w_1 is below the reservation wage w_r
 - The person will drop out of the labour force
- Now use all their time for leisure
 - Consume non-labour income Y_n

Moonlighting

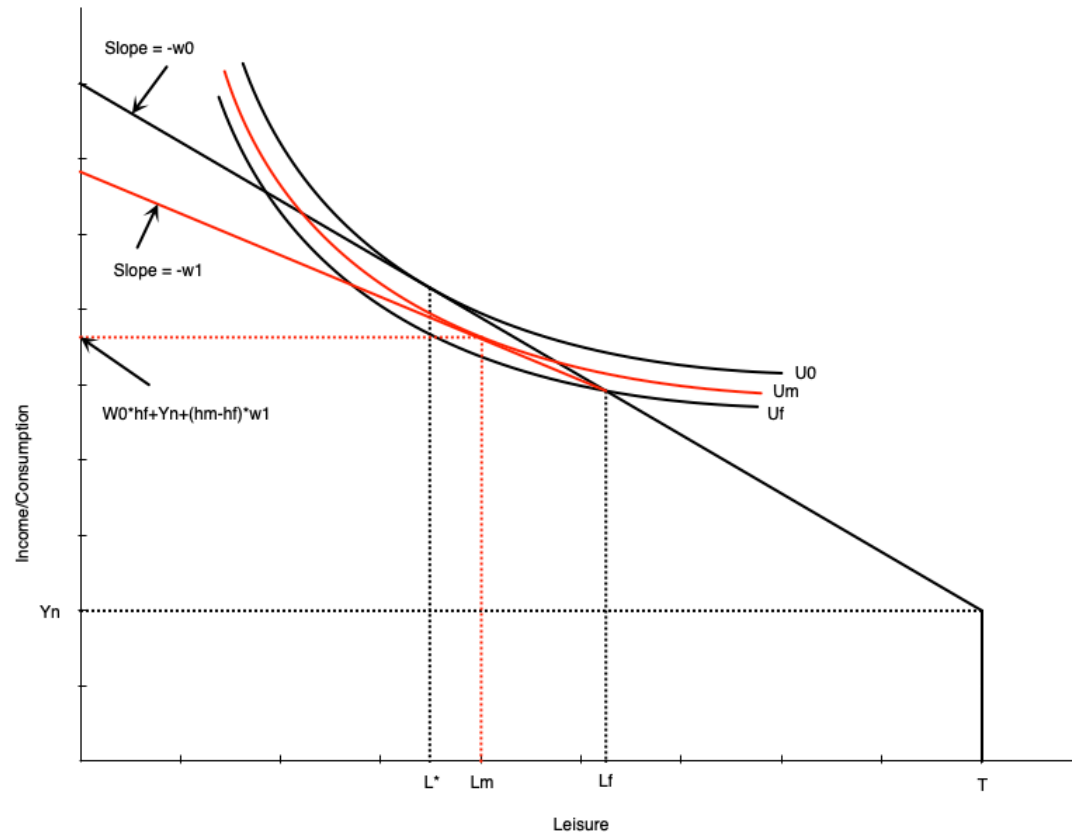
- **Moonlighting:** working additional jobs at night or on weekends
 - Bit of an old term where main jobs were generally 9-5
 - Extra work occurred after that
 - In 2025 secondary jobs can happen at any time
- Happens when people have a fixed hours constraint
 - Example: working full time, but wishing to work longer
 - People with fixed hours can be **underemployed**
- People may take a second job, possibly at a lower wage

Moonlighting



- In this graph people ideally would work $h^* = T - L^*$ hours
 - Maximizes utility subject to budget constraint
- Can only work $h_f = T - L_f$ hours at their main job
 - Employer does not allow more hours
- This puts them on a lower indifference curve U_f
- The outcome is not optimal

Moonlighting

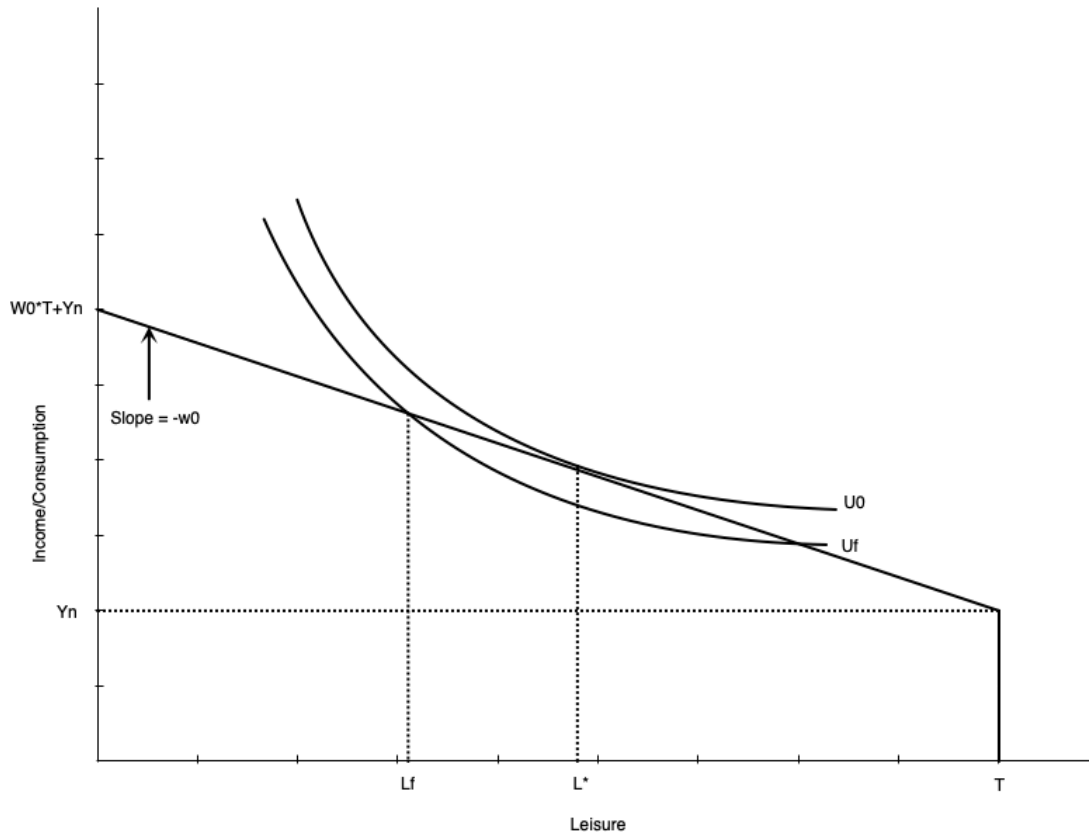


- Now imagine they can take a second job at a lower wage rate
- The budget constraint pivots out from the fixed hours point
- Beyond those hours, the wage falls
- Allows worker to access higher utility curve
 - They can increase hours worked
 - But not as much as if they could extend hours in main job

Overtime

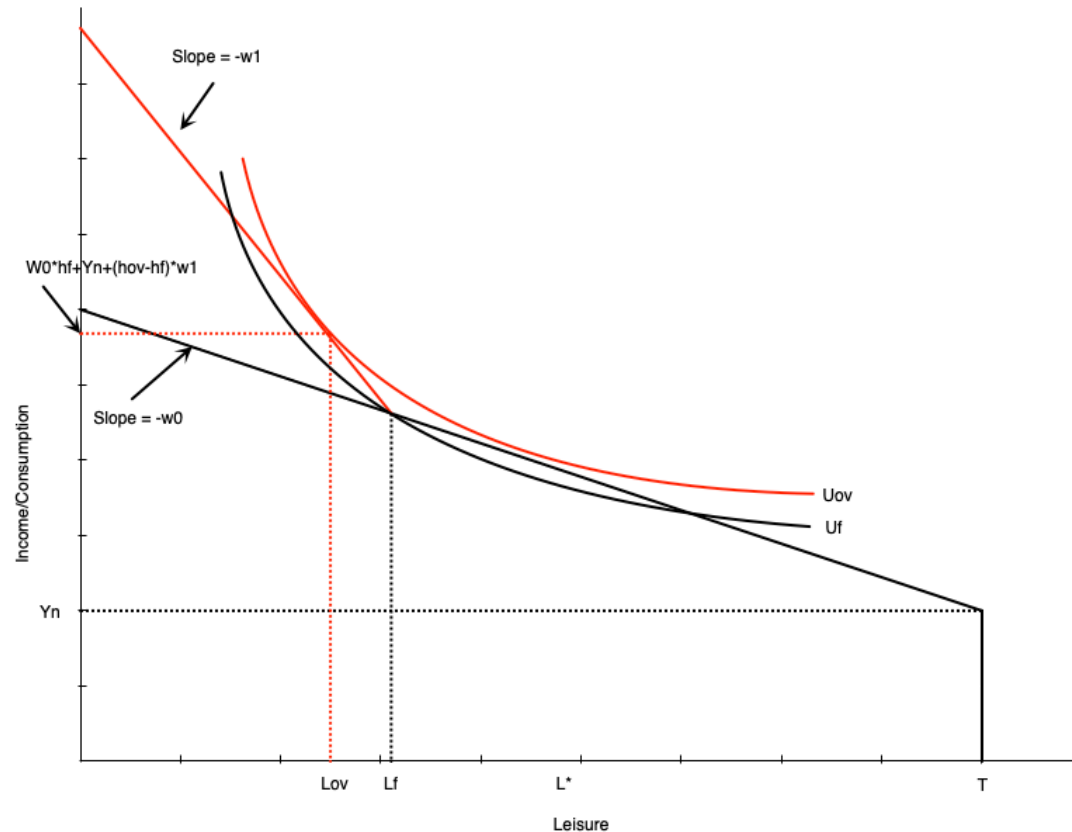
- Sometimes people are working too much at the main job
- Also happens with a fixed hours constraint
 - People want to work less at given wage
 - Employer does not allow less hours
 - They are **overemployed**
- At given wage they cannot maximize utility subject to constraints

Overtime



- In this graph, people would ideally work $h^* = T - L^*$ hours
 - Maximizes utility subject to budget constraint
- But employer requires them to work $h_f = T - L_f$ hours
 - Employer does not allow fewer hours
- Puts them on a lower indifference curve U_f
- The outcome is not optimal

Overtime

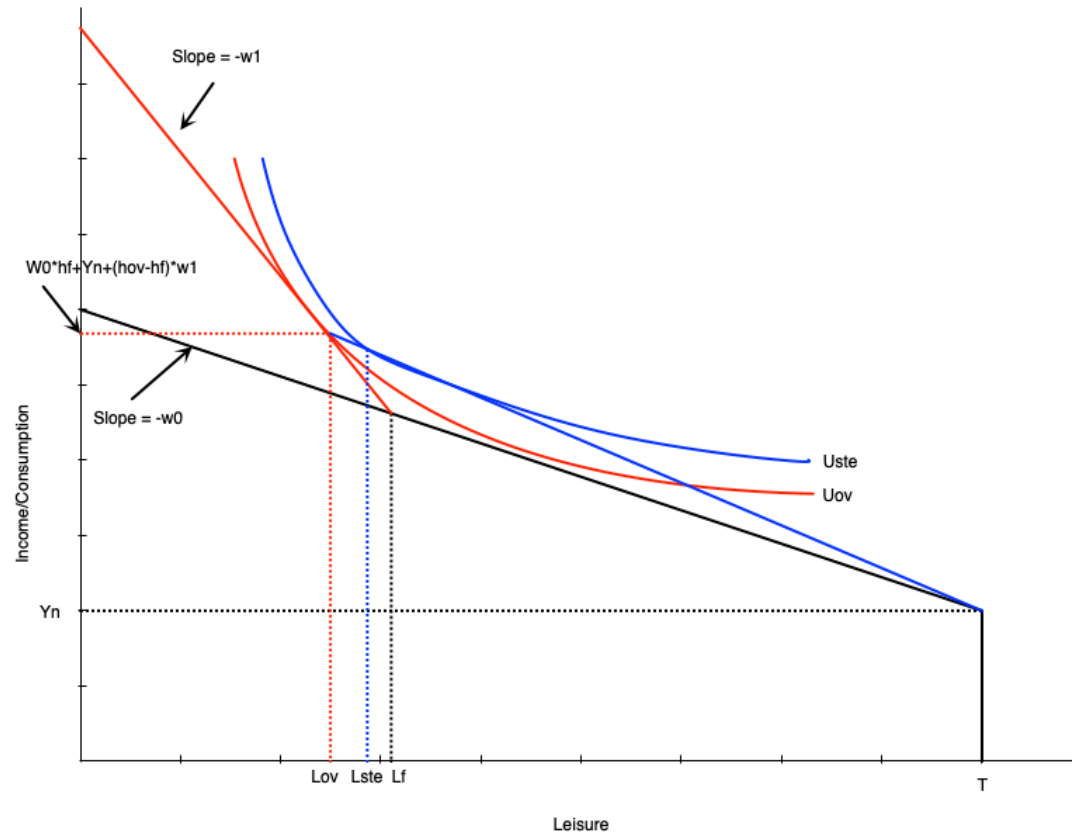


- If the employer offers overtime wages, the budget constraint pivots out from the fixed hours point
 - Beyond those hours, the wage increases from w_0 to w_1
- Allows worker to access higher utility curve
- Also increases hours worked even though they are initially overemployed
 - The higher wage induces a substitution effect to work more
 - There is a small income effect that only applies to the extra hours

Overtime

- An alternative to overtime is for the firm to just pay more for all hours
 - Makes sense for a firm if they are constantly paying overtime
- That does not generate the same optimal work hours as overtime pay
 - Income effect is larger with straight time equivalent
 - Because wage increase applies to *all* hours worked
 - In overtime, income effect only applies to extra hours worked
- Worker chooses to less with straight time equivalent pay

Overtime



- Straight-time equivalent work is depicted to the left
- The blue line goes from the fixed hours point with slope w_1
 - Generates the equivalent income as overtime pay at the optimum
- Worker chooses fewer work hours than overtime
 - Due to income effect of having higher wage at all work hours

Summary

Summary

- We have explored the labour-leisure model for labour supply
- The model is based on utility maximization subject to a budget constraint
- The model can explain both participation and work hours decisions
- Changes in non-labour income and the wage affect work decisions
- The labour supply curve can be backward bending
- The model can be used to analyze various situations faced by workers
 - Added worker effects
 - Discouraged workers
 - Moonlighting
 - Overtime
- Next chapter we will analyze government policy within this model